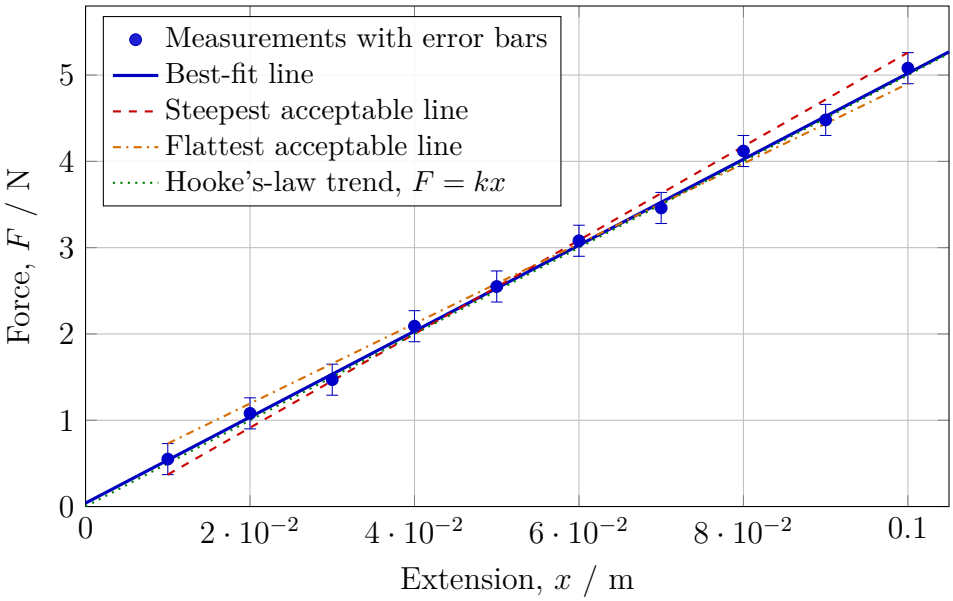
Relative / fractional (percentage) uncertainty: $\dfrac{\Delta x}{x}(\times 100\%)$

$$\frac{\Delta(x/y)}{x/y} \Big/ \frac{\Delta(xy)}{xy} = \frac{\Delta x}{x} + \frac{\Delta y}{y}$$

$$\frac{\Delta(kx^n)}{kx^n} = n \times \frac{\Delta x}{x}$$

0.5 Uncertainty in graph

Include error bars in the graph. Example of Hooke's Law applied with uncertainties:



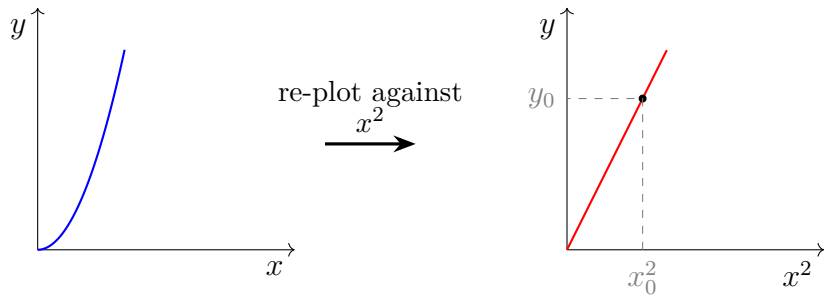
Worst acceptable line / error line (flattest: straight line passing through the highest point in first error bar and the lowest point in the last error bar; steepest: straight line passing through the lowest point in first error bar and the highest point in the last error bar)

Gradient (including uncertainties): $m \pm \dfrac{|m_1 - m_2|}{2}$

Intercept (including uncertainties): $c \pm \dfrac{|c_1 - c_2|}{2}$

0.6 Linearization of graph

- Non-logarithmic approach: transform the relationship into linear relationship via changing the x-axis being graphed against (e.g. $y \propto x^2$) This is learnt in Additional Mathematics, so it should be a revisit.
- Logarithmic approach: transform the relationship into linear ones via applying logarithmic functions (Example: $y = ax^b \Rightarrow \log y = b \log x + \log a$; Example 2: $y = ae^{bx} \Rightarrow \ln y = \ln a + bx$)



Original: $y = 2x^2$ (curve) Linearized: $y = 2x^2$, gradient $= 2$

1.3 Graphs of motion

For any motion graph:

$$v = \frac{ds}{dt}, \qquad a = \frac{dv}{dt} = \frac{d^2s}{dt^2}$$

and the area under a $v - t$ graph gives displacement, while the area under an $a - t$ graph gives change in velocity.

Take $x = 2\pi t$, where $0 \leq x \leq 2\pi$ corresponds to $0 \leq t \leq 1.0\,\text{s}$. If

$$s(t) = \sin(2\pi t),$$

then

$$v(t) = \frac{ds}{dt} = 2\pi\cos(2\pi t), \qquad a(t) = \frac{dv}{dt} = -(2\pi)^2\sin(2\pi t).$$

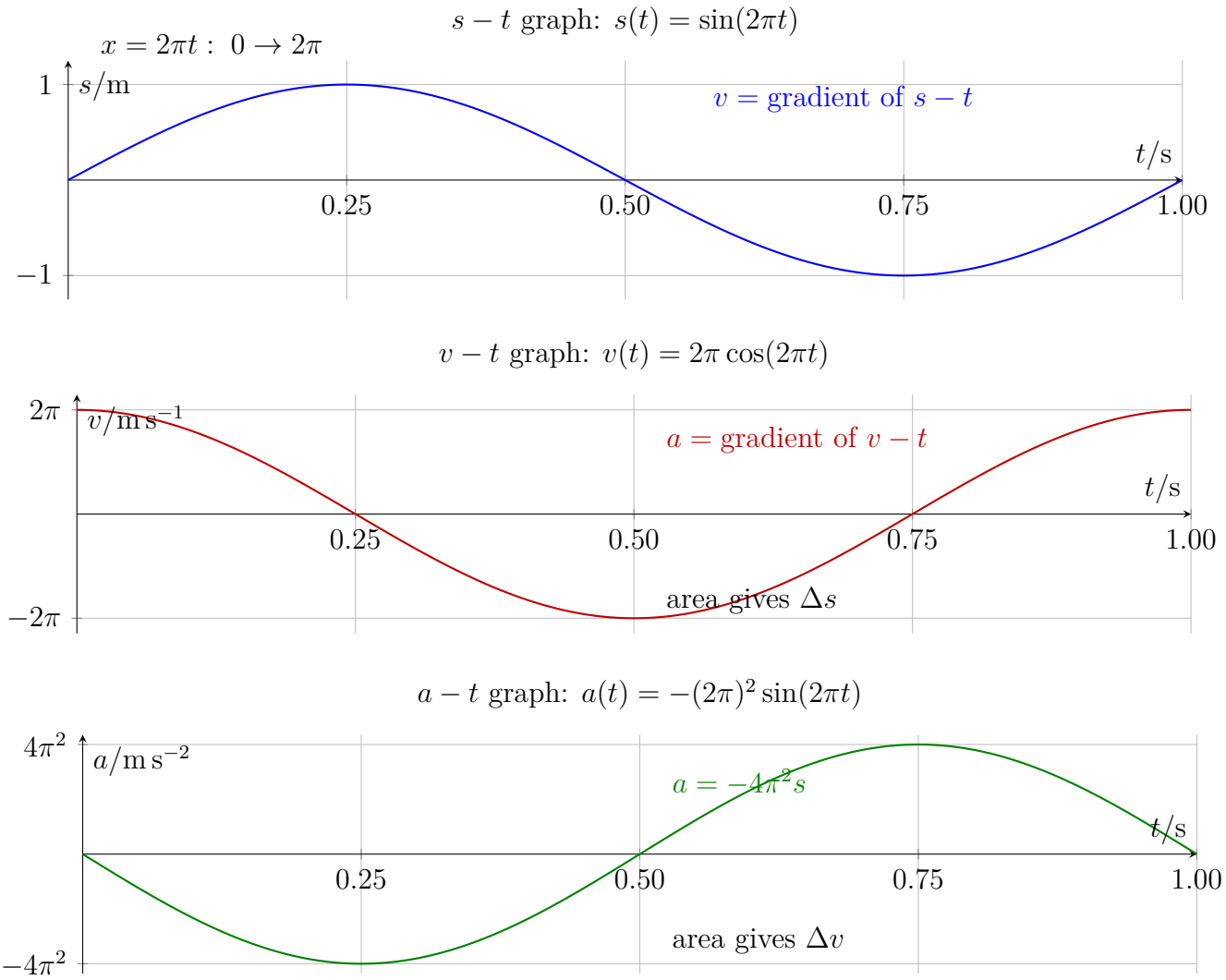


Figure 1: The $s - t$, $v - t$, and $a - t$ graphs for one cycle of sinusoidal motion.

1.4 Projectile motion

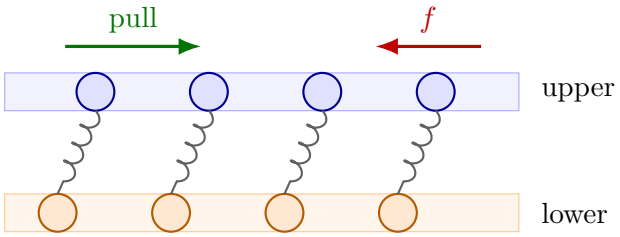
Types of projectiles:

- Up and down

- $F_B = \rho g V_{\text{imm}}$, where V_{imm} is the volume of the body that is immersed in the fluid.
- Caused by the pressure that the fluid exerts on the body.
- 王忠 Let's go: $F_{浮} = \rho_{液} g V_{排}$

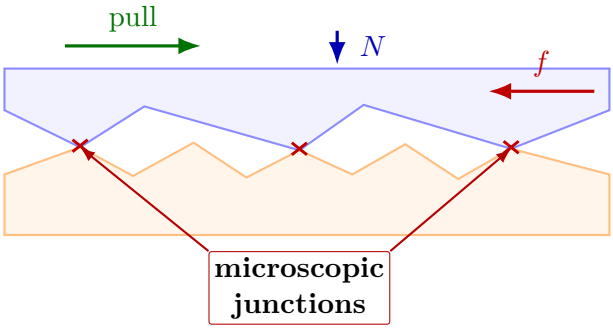
- Frictional forces
  - Generally oppose the motion of a body, electrical nature, parallel to contact surface
  - Occurs when one body slides over another / has a tendency for motion, has dynamic / static friction

Simple atoms-and-springs model



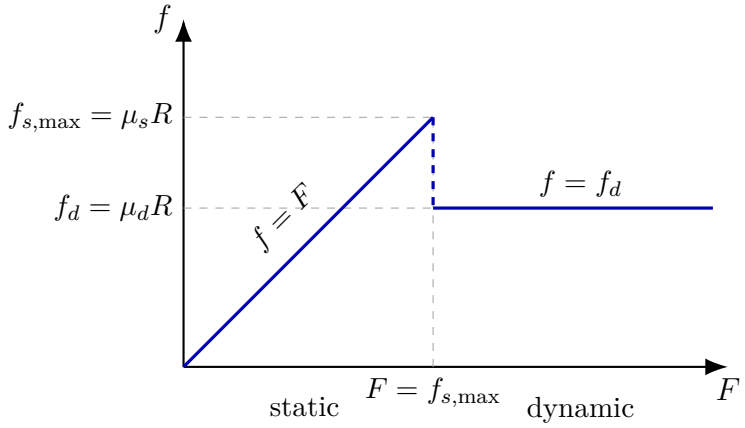
Atoms are displaced; stretched microscopic bonds act like springs and push back.

Asperities model



Real surfaces touch at asperities; junctions must shear or break before sliding occurs.

  - Friction Laws:
    1. Area of contact does not affect the frictional force.
    2. The force of dynamic friction is $f_d = \mu_d R$, where R is the normal force and μ_d is the coefficient of dynamic friction.
    3. The force of dynamic friction does not depend on the speed of sliding.
    4. The MAXimum force of static friction is $f_s = \mu_s R$, where μ_s is the coefficient of static friction and $\mu_s > \mu_d$
    5. ($\mu_s R$ does not give the force of static friction, but only the MAXimum)
    6. $0 \leq F_s \leq F_{s,\max}$; $F_d < F_{s,\max}$; $F_d = k$



2.3 Circular motion

Circular motion requires the presence of a force directed towards the center of the circle.

$$v = \frac{2\pi r}{T} = r\omega = 2\pi r f$$

where $T = \dfrac{2\pi r}{v} = \dfrac{2\pi}{\omega}$ is the period / time taken for one complete revolution.

$$\text{Angular speed} = \omega = \frac{\text{angle swept}}{\text{time taken}} = \frac{\Delta\theta}{\Delta t}$$

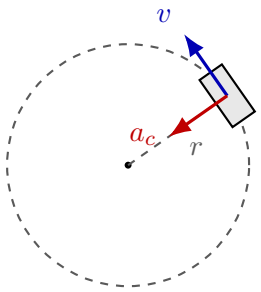
For a complete revolution, $\omega = \dfrac{2\pi}{T} = 2\pi f$.

Frequency, the number of complete revolutions in one second: $f = \dfrac{1}{T}$.

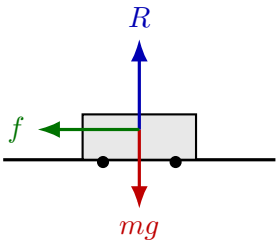
Centripetal acceleration, normal to the circle and directed toward the center of the circle:

$$a_c = v\omega = \frac{v^2}{r} = \omega^2 r = \frac{4\pi^2 r}{T^2} = 4\pi^2 r f^2$$

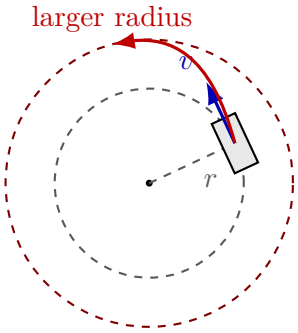
The centripetal force is directed towards the center of the circle (the direction of the net force).
A car will skid outward (i.e., will cover a circle of larger radius) if the friction force is not large enough.



top view

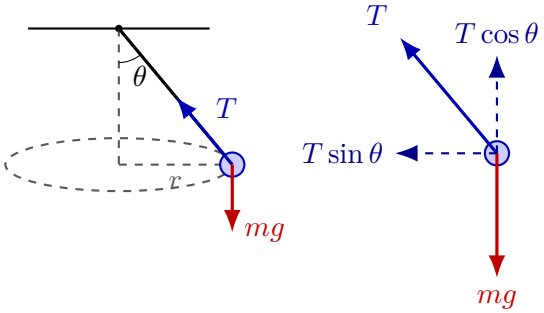


side view



speed too high

Question: A particle is tied to a string and moves with constant speed in a horizontal circle. The string is tied to the ceiling. Draw the forces on the particle.
The common mistake here is to put a horizontal force pointing toward the center and call it the centripetal force. But in fact, it already exists (the horizontal component of the tension).



The steps to solve problems on circular motion:

1. Analyze the state of motion of the studied object (if it is in a state of uniform circular motion).
2. Determine the center and the radius of the circle.
3. Analyze the forces, determine the origin of centripetal force, and calculate the resultant force in the direction to the center of the circle.
4. The resultant force provides the centripetal force of the circular motion.

4.2 Impulse and force-time graphs

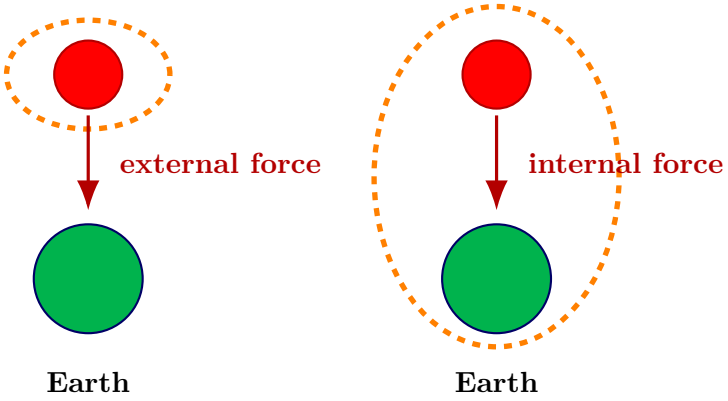
$$\vec{J} = \Delta\vec{p} = \vec{F_{\text{net}}}\Delta t$$

In other words, impulse is the area under the curve in a graph of force against time.

4.3 Conservation of momentum

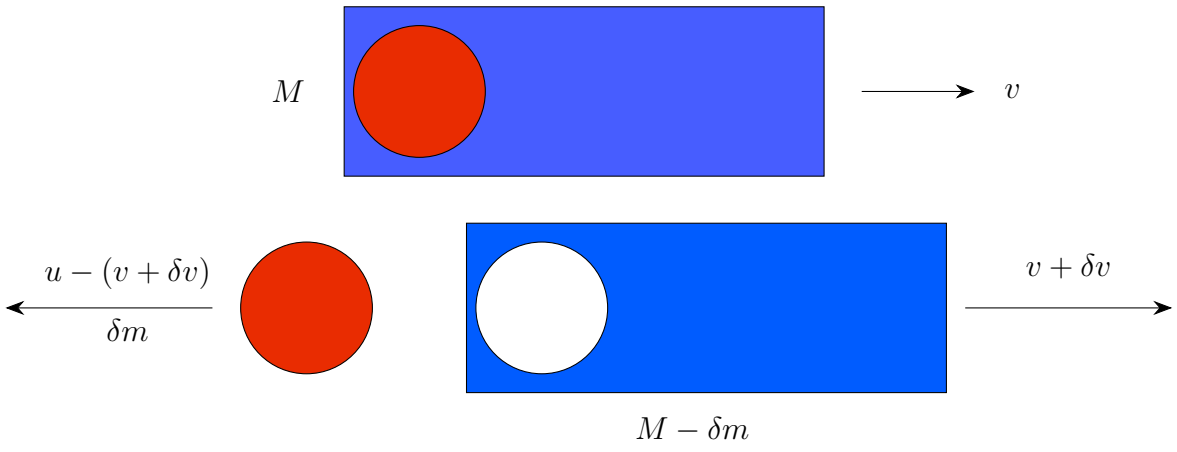
Conservation of momentum: when the net force on a system is zero the momentum does not change and it is conserved. The total momentum of a closed system of colliding objects remains constant in any direction iff no external forces act on the system.

Consider a ball that you drop from rest from a certain height. As the ball falls, its speed and momentum increase so the momentum of the ball does not stay the same. This is to be expected—there is an external force on the ball, namely its weight. So the momentum of the system that consists of just the falling ball is not conserved. If we include the earth as part of the system, then there are no external forces anymore and the total momentum will be conserved.



Extension: rocket equation 火箭方程

The rocket moves with speed v. The engine is turned on, and gases leave the rocket with speed u relative to the rocket. The initial mass of the rocket including the fuel is M. After a short time δt the rocket has ejected fuel of mass δm. The mass of the rocket is, therefore, reduced to $M - \delta m$ and its speed increased to $v + \delta v$.



5.3 Angular momentum

(A vector) It is defined as

$$L = I\omega \Rightarrow \tau_{\text{net}} = \frac{\Delta L}{\Delta t}$$

Conservation of angular momentum: When the net torque on a system is zero, the angular momentum is conserved; that is, it stays constant.

$$J = \Delta L = \tau_{\text{net}} \Delta t$$

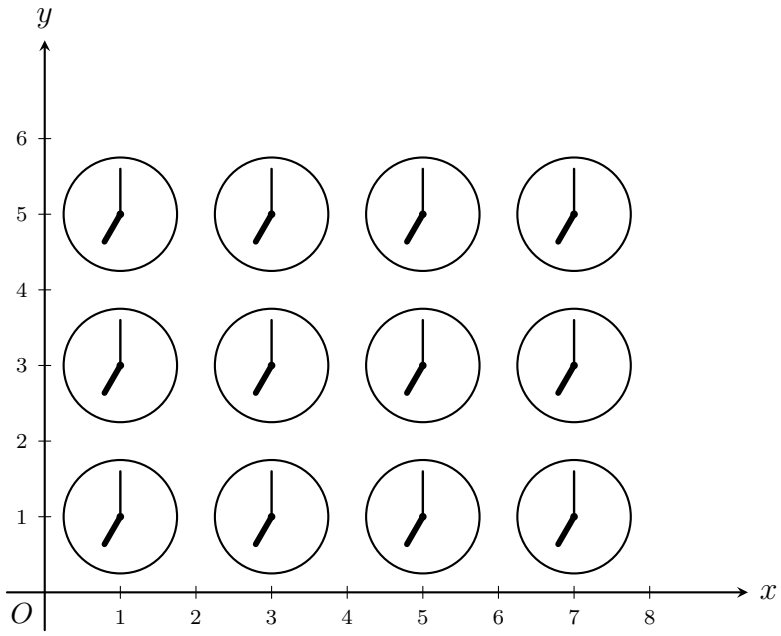
$$E_k = \frac{L^2}{2I}$$

6 Relativity

可以这么说：一切都是相对的，没有一个东西是绝对地“在动”，也没有一个东西是绝对的静止。

6.1 Reference frames and Lorentz transformations

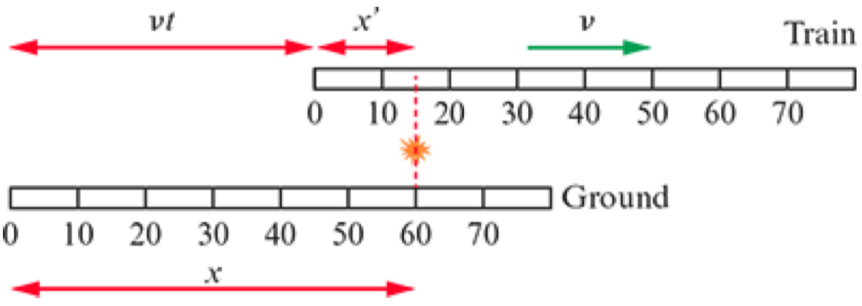
A reference frame is a set of coordinate axes and a set of clocks at every point in space. If this set is not accelerating, the frame is called an inertial reference frame.



Galilean transformation equations:

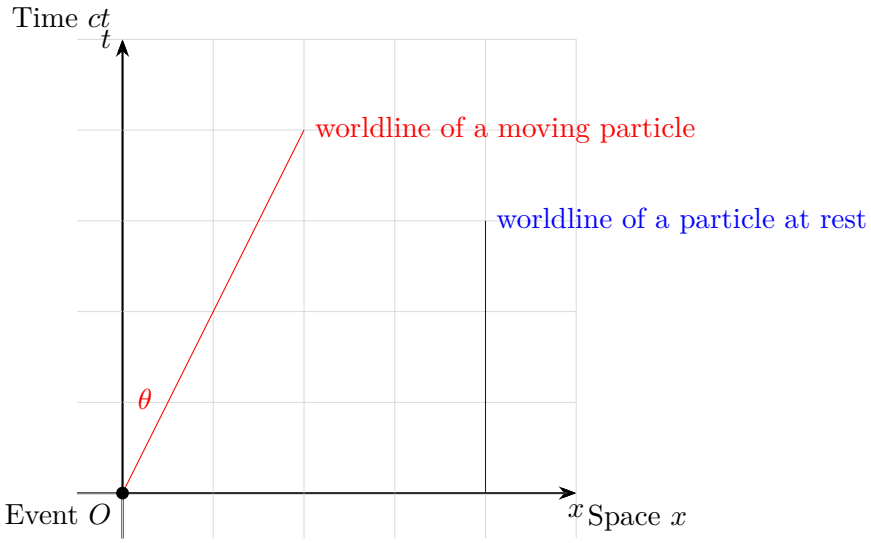
$$x' = x - vt;\ t' = t;\ u' = u - v$$

where x and x' denotes the position that the event occurred to two observers.

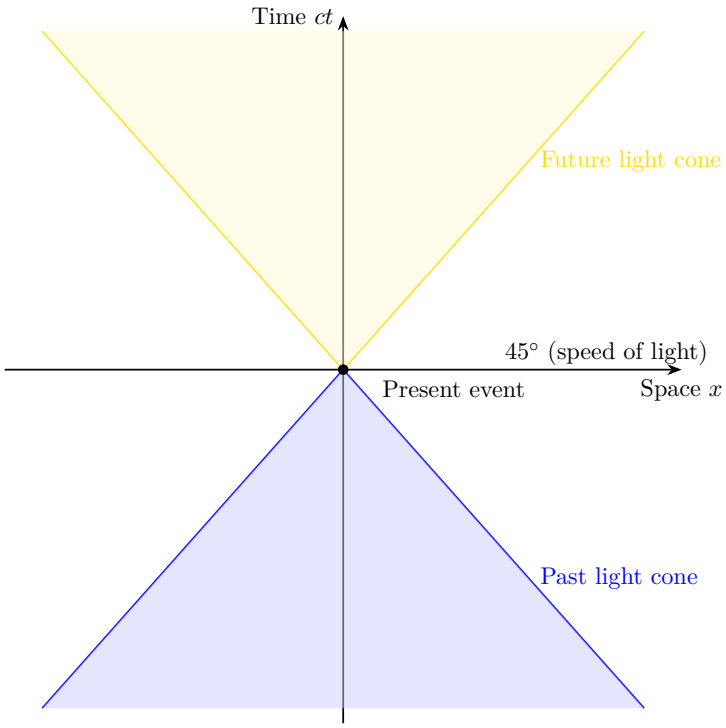


6.3 Spacetime / Minkowski diagrams

Do not be led astray by your knowledge of Euclidean geometry!
Time is the y-axis (plotted as ct though, in meters), distance (1D) is the x-axis.



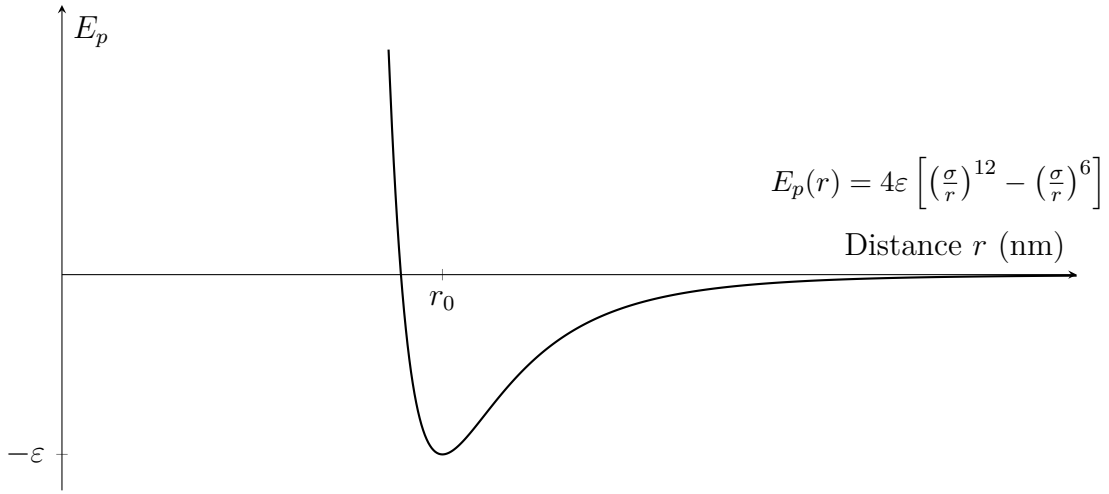
The tangent of the angle between the worldline and the ct axis gives the speed divided by the speed of light. Since nothing can exceed c, the worldline of any particle will always make an angle less than 45° with the ct axis.



The region coloured yellow consists of all events that can be influenced by event E. Similarly, events that could have influenced E are within the region coloured blue.

重要概念：其实坐标轴可以是斜的。想象有一个 S' 以 v 的速度向右移动，且 S 和 S' 的原点重合。ct' 的坐标轴会和 ct 坐标轴形成 $\theta = \tan^{-1}\left(\frac{v}{c}\right)$ 的夹角。同样地，x' 的坐标轴会和 x 坐标轴形成 θ 的夹角。所有坐标轴均对称于光子的世界线（因为光速在所有的 Frames 里的速度都是一致的）。

Use the gamma factor and Lorentz transformation to find x and ct from x' and ct' values. ($x = \gamma(x' + vt')$, $ct = \gamma(ct' + vcx')$) 当 x 和 ct 等于伽马系数的时候，向右/向上延伸与 ct' 和 x' 相交的点就是 S' 下的 1m。



7.2 Specific heat capacity and change of phase

Specific heat capacity: the energy required to increase the temperature of a unit mass of the body by one kelvin.

$$Q = mc\Delta t$$

where Q is the amount of heat needed, m is the mass, Δt is the change in temperature.

Changes of phases:

- melting: when a solid changes to a liquid (thermal energy is transferred to the solid)
- freezing: when a liquid changes into a solid (thermal energy is transferred away from the liquid)
- vaporisation (or boiling): when a liquid changes into vapour (thermal energy is transferred to the liquid)
- condensation: when a vapour changes into a liquid (thermal energy is transferred away from the vapour).
- evaporation: vaporisation taking place at the surface of a liquid at any temperature (rate increases with higher surface area and higher temperature)

Latent heat: The energy supplied or removed during a phase change (does not increase / decrease the temperature of the substance)
Specific latent heat: the energy required to change the phase of 1 kg of a substance at constant temperature and pressure.

$$Q = mL$$

where m is the mass, Q is the energy applied, L_F is the **specific latent heat of fusion** (for melting / freezing), L_V is the **specific latent heat of vaporization** (for vaporization / condensation)
Generally speaking, $L_V > L_F$.

What happens during change in phase:

- The thermal energy supplied increases the intermolecular potential energy, internal energy increases
- This work increases the separation of the molecules
- The average distance between the molecules increases
- Change in phase occurred

The transfer of thermal energy between (melted) water and ice ensures that the overall equilibrium temperature stays at 0°C.

Two methods for measuring specific heat capacity:

- Electrical method
  1. Record initial temperature of the substance
  2. Apply heat from electrical heater (with known power / voltage + current) for a certain amount of time
  3. Measure the final temperature, calculate specific heat capacity
- Method of mixtures
  1. A solid is put in a container of hot water and allowed time to reach a constant temperature
  2. The temperature of the solid is recorded (equal to temperature of water)
  3. The solid is then transferred into an insulated calorimeter 热量计 of known specific heat capacity and initial temperature which contains a liquid
  4. Final temperature of water is then recorded (when thermal equilibrium reached)
  5. $$\underbrace{Q}_{\text{substance with unknown } c_{sp}} = \underbrace{Q}_{\text{calorimeter}} + \underbrace{Q}_{\text{liquid (water)}}$$

     L_F/L_V could also be calculated this way:

     $$\underbrace{Q}_{\text{latent heat}} + \underbrace{Q}_{\text{heat for change in temperature}} = \underbrace{Q}_{\text{calorimeter}} + \underbrace{Q}_{\text{liquid (water)}}$$
  6. (It is likely that some thermal energy (heat) was dissipated to the surrounding air, therefore the real c_{sp} might be larger than the calculated value)

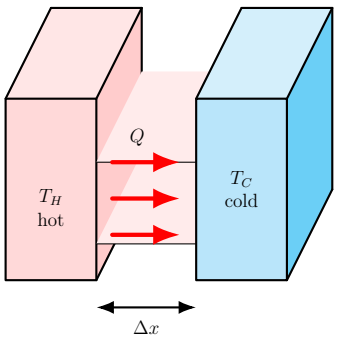
The rate of cooling is proportional to the difference in temperature between the liquid and the surroundings:

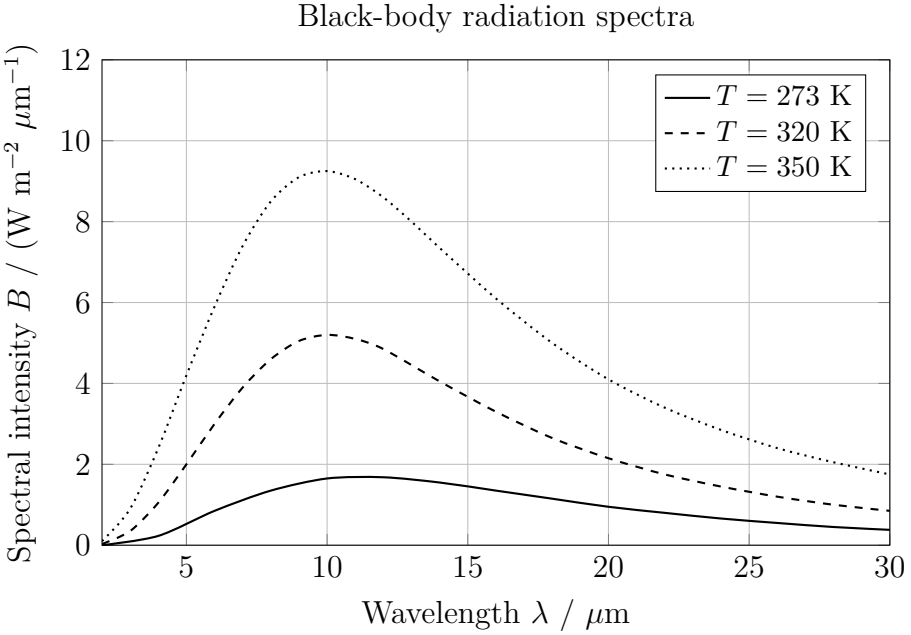
$$\frac{\Delta T}{\Delta t} = k(T - T_s)$$

where the constant k is proportional to the area of the surface of the liquid.
(This implies an exponential decrease of temperature)

7.3 Thermal energy transfer

- Conduction
  - The moving electrons collide with neighbouring molecules, transferring energy to them and so increasing their average kinetic energy.





* : Spectral intensity: power emitted per unit area per unit wavelength

The area under the graph is the radiated intensity. With increasing temperature, the peak of the curve occurs at lower wavelengths and the height of the peak increases. The energy radiated is electromagnetic radiation and is distributed over an infinite range of wavelengths. The peak of the spectrum corresponds to a specific wavelength $\lambda_{\max}$, called the peak wavelength.

Wien's Law:

$$\lambda_{\max} T = 2.90 \times 10^{-3}\ \mathrm{K\,m}$$

This shows that the peak wavelength λ_0 depends on temperature.

Apparent brightness (the power received per unit area of the detector):

$$B = \frac{L}{4\pi d^2} = \frac{\sigma A T^4}{4\pi d^2}$$

where L is the luminosity of the star, d is the distance from the star to the observer.
Luminosity of a star is given by

$$L = \sigma A T^4$$

where A is the surface area of the star, T is the temperature of the star.

8 The greenhouse effect

8.1 Radiation from real bodies

Real bodies radiate according to the modified Stefan-Boltzmann law:

$$P = e\sigma A T^4$$

where e is the emissivity of the body, a number between 0 and 1, the ratio of the power per unit area radiated by a body to the power per unit area radiated by a black body of the same temperature. A perfect black body has an emissivity of 1, while a perfect reflector has an emissivity of 0.
The power radiated per unit area is called (radiated) intensity, and is equal to $I = e\sigma T^4$.

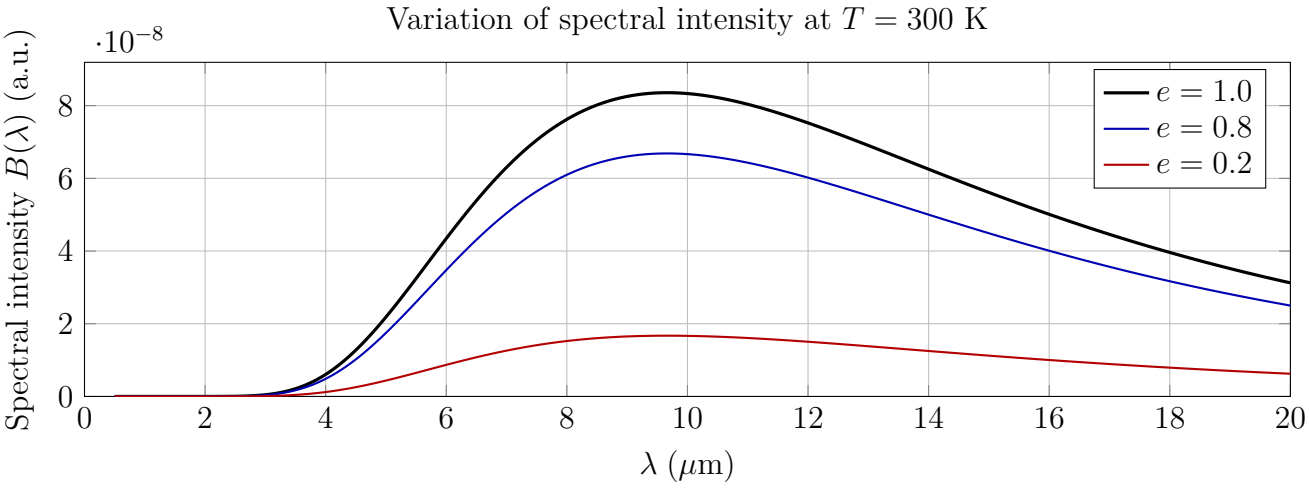
Albedo of the body is the ratio of the reflected intensity (or power) to the incident intensity (or power).

$$\alpha = \frac{\text{total reflected power}}{\text{total incident power}}$$

For a body that can only absorb or reflect, $e + \alpha = 1$.
If considering the temperature of the surrondings, T_S:

$$P_{\text{net}} = e\sigma A(T^4 - T_S^4)$$

$$I_{\text{net}} = e\sigma(T^4 - T_S^4)$$



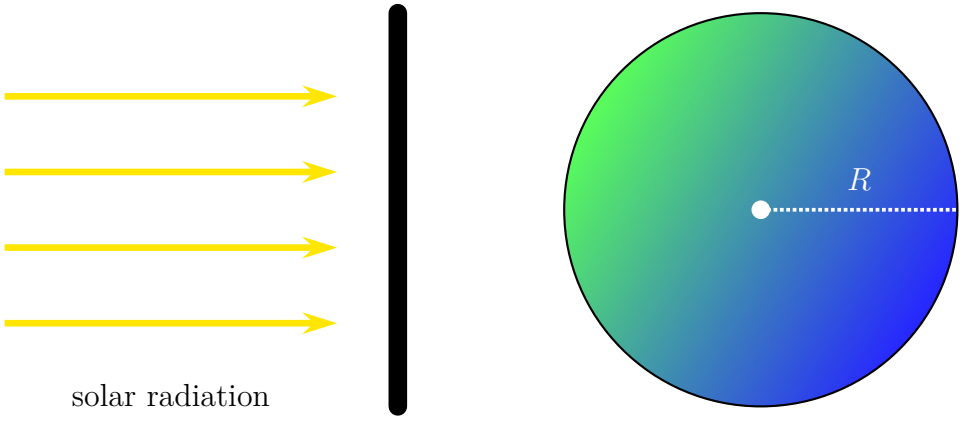
The solar constant:

$$S \approx 1400 \text{ W m}^{-2}$$

(calculated using $I = \frac{P}{4\pi d^2}$, where I is intensity, power of radiation received or emitted per unit area; $P = 3.9 \times 10^{26}$ W (total power the sun emitted), $d = 1.5 \times 10^{11}$ m $= 1$ au, average earth-sun distance)
It is the intensity of the solar radiation at the top of the earth's atmosphere.
The intensity emitted by the sun is

$$I = \frac{3.9 \times 10^{26} \text{ W}}{4\pi(7.0 \times 10^8 \text{ m})^2} = 6.3 \times 10^7 \text{ W m}^{-2}$$

8.2 Energy balance of the earth



disc of area $4\pi R^2$ surface area of the sphere $4\pi R^2$

The radiation that reaches the earth has to go through the area of a disc of radius R (the Earth's radius). The power going through the disc is given by

$$P = S\pi R^2 (= SA)$$

where S is the solar constant, A is the area of the disc.
The average intensity reaching the Earth's surface is

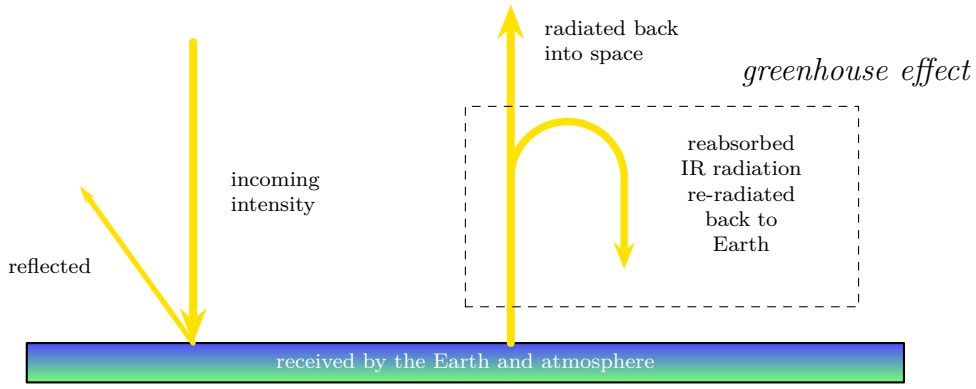
$$I_{\text{av}} = (1-\alpha)\frac{S}{4} = 245 \text{ W m}^{-2}$$

where $\alpha \approx 0.30$ (albedo of the earth varies and depends on (for example) the latitude and whether covered by clouds or not, here 0.3 is an average), and $\frac{S}{4}$ is found by diving the the power through the disc by the surface area of the earth.

Energy balance equation: (very simple, not considering greenhouse effect)

$$(1-\alpha)\frac{S}{4} = \sigma T^4$$

Greenhouse gases



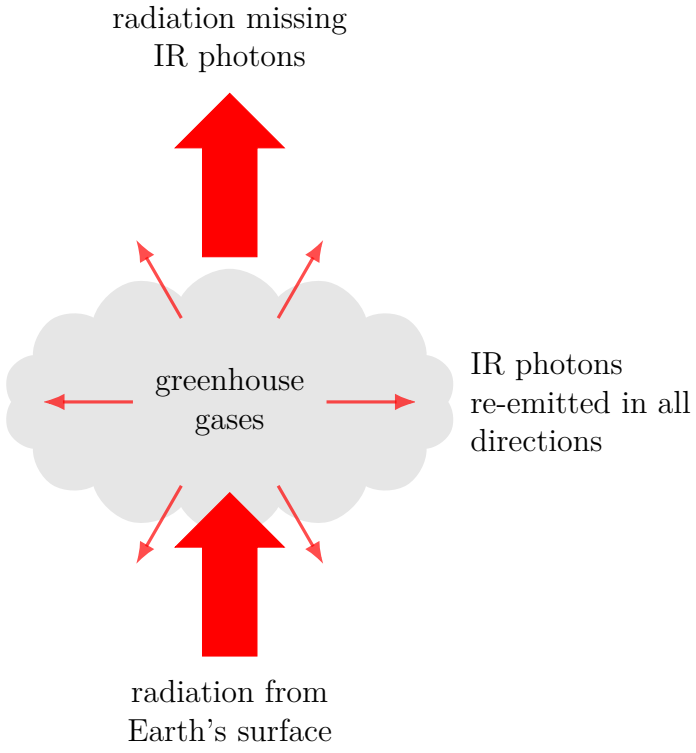
Earth's surface

A thorough process of how did the greenhouse effect occur:

1. Infrared radiation emitted from the Earth's surface is strongly absorbed by greenhouse gases in the atmosphere
2. This radiation is in turn re-radiated by these gases in all directions.
3. This means that some of this radiation is received by the earth's surface again, causing additional warming.

Enhanced greenhouse effect: additional warming due to increased quantities of the greenhouse gases in the atmosphere. The increases in the gas concentrations are due to human activity and are closely related to the burning of fossil fuels.

Greenhouse gas	Natural sources	Anthropogenic sources
H_2O	evaporation of water from oceans, rivers and lakes	irrigation
CO_2	forest fires, volcanic eruptions, evaporation of water from oceans	burning of fossil fuels, deforestation
CH_4	wetlands, ocean, lakes, rivers, termites	agriculture (rice cultivation), livestock farming, landfill sites
N_2O	oceans, forests, soils, tropical forests	agriculture (fertilizer use), burning of fossil fuels, munufacture of cement, deforestation (reduction of nitrogen fixation)



The energy of the molecules are **discrete**. The difference in energy between molecular energy levels is the energy of an infrared photon.

- Infrared photons travelling through greenhouse gases will be absorbed.
- The gas molecules that have absorbed the photons will now be excited to higher energy levels.
- But the molecules prefer to be in low-energy states, and so they will immediately make a transition to a lower-energy state by emitting the photons they absorbed.
- The emitted photons have the same energy as the absorbed photons, but they are emitted in random directions.
- Some are emitted back towards the earth, thereby warming the earth's surface.

OR

- **Resonance:** All systems have their own natural frequency of oscillation, and resonance happens when the frequency of the incoming radiation matches the natural frequency of the system.

- In the case of the greenhouse gases, the natural frequency is the frequency of the photons the greenhouse gases can absorb.
- Resonance is achieved when the photons emitted by the earth's surface have the same frequency as the natural frequency. (i.e., the energy of the photons emitted by the earth's surface is the same as the energy difference between the molecular energy levels of the greenhouse gases)
- Photons with the right energy (frequency) will be absorbed by the greenhouse gases, and then re-emitted in random directions, some of which will be back towards the earth's surface, causing additional warming.

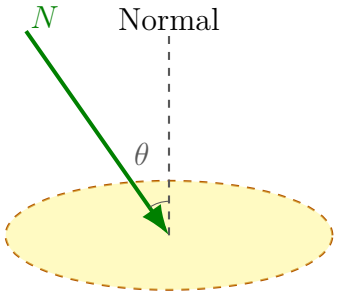
9 The gas laws

9.1 Moles, molar mass and the Avogadro constant

I assume we have some IG chemistry memory coming back?
1 mole of any substance contains as many particles (atoms / molecules, depends) as atoms in 12 g of carbon-12.
1 mol $= 6.02 \times 10^{23}$ particles
This is also known as the Avogadro constant, N_A.
$n = \dfrac{N}{N_A}$ if n is the number of moles, N is the number of particles.
The molar mass of a substance is the mass (in grams) of 1 mole of that substance.
${}^{A}_{Z}X$ has approximately a molar mass of A g/mol.
$n = \dfrac{m}{\mu}$, where μ is the molar mass, m is the mass of the substance, n is the number of moles.

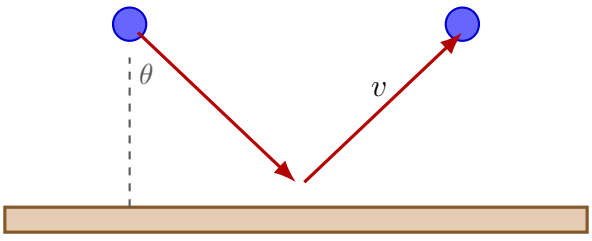
9.2 Ideal gases

Pressure: the normal force applied per unit area. Commonly known as $P = \dfrac{F}{A}$, but as shown in the diagram below, the formula should be adjusted as $P = \dfrac{F\cos\theta}{A}$.



1 atm $\approx 1.01 \times 10^5$ Pa

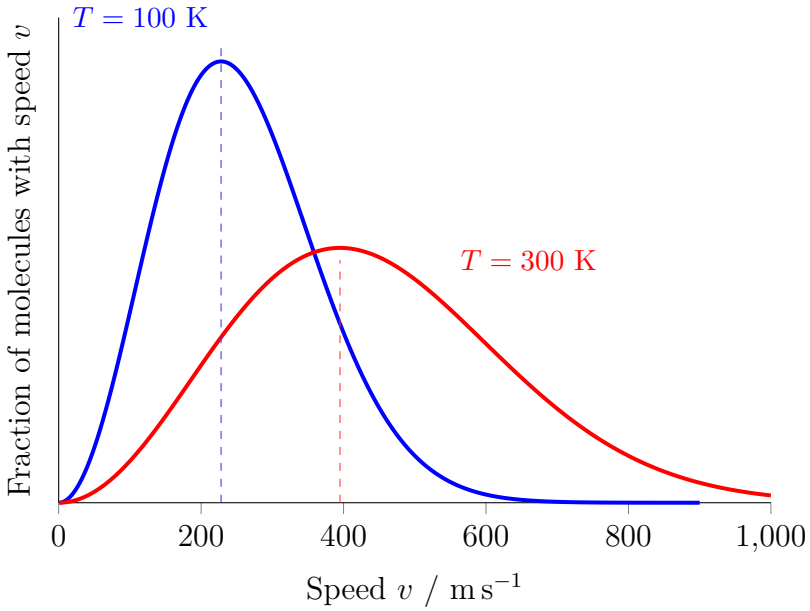
Lydia 丝滑连招解释压强（哎呀 hào sí huà yā）：



Container wall

9.3 The Boltzmann equation

The molecules of a gas move about randomly with a range of speeds. The speed at the peak represents the most probable speed that would be found if you picked a molecule at random.



The root mean square speed of the molecules is given by

$$c = \sqrt{\frac{\sum_{i=1}^{n} v_i^2}{N}}$$

The average kinetic energy of the molecules is given by

$$\overline{E}_k = \frac{1}{2}mc^2$$

where m is the mass of a molecule, c is the root mean square speed of the molecules.
Note that we will use c to represent the root mean square speed, most probable speed, and the average speed.
The equation for the pressure of the gas is given by

$$P = \frac{1}{3}\rho c^2$$

下面是推导过程（考试不需要）：
$\vec{J} = 2mv_x$ (a molecule with mass m and velocity v_x collides with the wall and rebounds)
$T = 2\dfrac{L}{v}$ (time between collisions on the right side of the wall, assume the length of the container is L)
$\dfrac{J}{T} = \dfrac{2mv_x}{2\frac{L}{v}} = \dfrac{mv_x^2}{L}$ (average force exerted by the molecule on the wall over large amount of time)
$P_{\text{molecule}} = \dfrac{mv_x^2}{\frac{L^3}{N}} = \dfrac{mv_x^2}{V}$ (pressure exerted by the molecule on the wall, container is a cube)
$P_{\text{total}} = \dfrac{m}{V}\sum_{i=1}^{N} v_{x,i}^2$ (pressure exerted by all the molecules on the wall)
$\overline{v_x^2} = \dfrac{\sum_{i=1}^{N} v_{x,i}^2}{N}$ (average of v_x^2) $\Rightarrow P = \dfrac{mN}{V}\overline{v_x^2}$
$\because \overline{v_x^2} = \overline{v_y^2} = \overline{v_z^2}$ and $\overline{v_x^2} + \overline{v_y^2} + \overline{v_z^2} = c^2 \therefore \overline{v_x^2} = \dfrac{c^2}{3} \Rightarrow P = \dfrac{1}{3}\dfrac{mN}{V}c^2 = \dfrac{1}{3}\rho c^2$

The average kinetic energy of the molecules is also given by

$$\overline{E}_k = \frac{3}{2}k_B T$$

where k_B is the Boltzmann constant, T is the absolute temperature of the gas.
This exceptional equation relates a microscopic quantity, the average kinetic energy of molecules, to a macroscopic quantity, temperature. It also says that at $T = 0$, all molecular motion stops.

The total internal energy of an ideal gas (i.e. total random kinetic energy) is given by

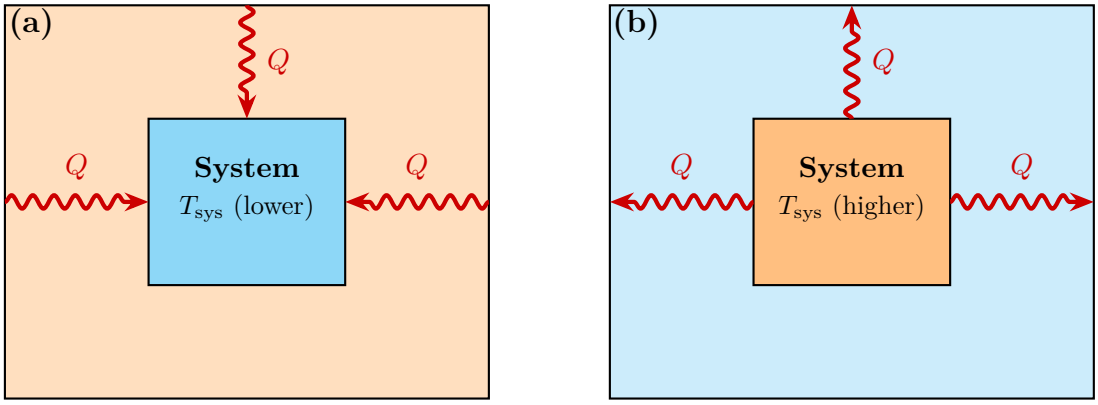
$$U = \frac{3}{2}Nk_B T = \frac{3}{2}nRT = \frac{3}{2}PV$$

10 Thermodynamics

10.1 Internal energy

The change in internal energy: (ideal gas)

$$\Delta U = \frac{3}{2}Nk_B \Delta T = \frac{3}{2}nR\Delta T = \frac{3}{2}P\Delta V = \frac{3}{2}V\Delta P$$



$T_{sys} < T_{surr}$
Heat *enters* the system

$T_{sys} > T_{surr}$
Heat *leaves* the system

Concepts regarding *systems*:

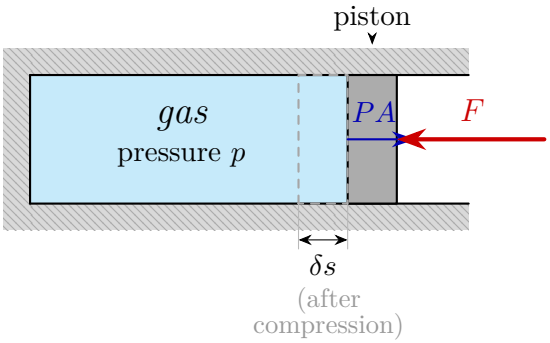
- Systems: the complete set of objects under consideration
- Closed system: a system that can exchange energy but not matter with its surroundings, mass can not enter or leave the system.
- Isolated system: a system that can exchange neither energy nor matter with its surroundings
- If all the parameters defining the system are given, we speak of the system being in a particular state
- Any process that changes the state of a system is called a thermodynamic process

Internal energy is a property of the particular state of the system under consideration and therefore is a state function.Heat and work are related to changes in the state of the system, not to the state itself.

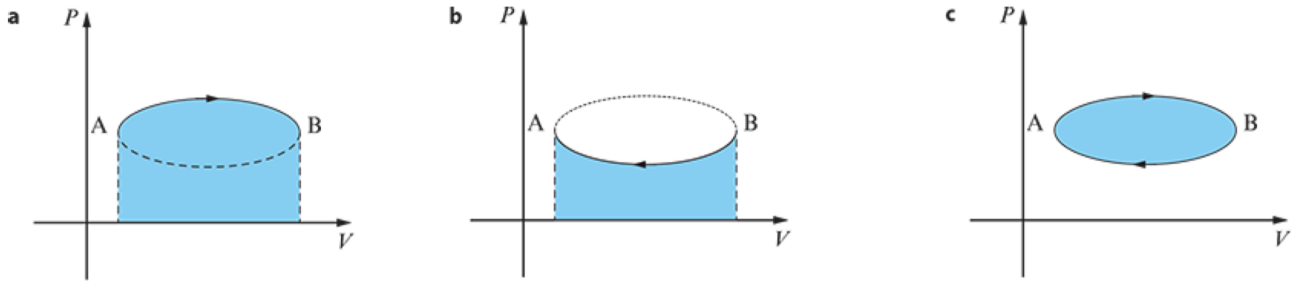
Process	Description	Equation
Isothermal	Temperature is constant	$P_1V_1 = P_2V_2$
Isobaric	Pressure is constant	$\frac{V_1}{T_1} = \frac{V_2}{T_2}$
Isovolumetric	Volume is constant	$\frac{P_1}{T_1} = \frac{P_2}{T_2}$
Adiabatic	No heat exchange	$P_1V_1^{\gamma} = P_2V_2^{\gamma}$, $\gamma = \frac{5}{3}$ $T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$, $\gamma - 1 = \frac{2}{3}$

Work done on / by the gas:

$$W = F\Delta s = PA\Delta s = P\Delta V$$



It could also be calculated by finding the area under the curve in a P-V graph. (when pressure is not constant)



For a closed loop in a pressure-volume diagram, the work done is the area of the loop.
(Work done expanding is shown in a, work done compressing is shown in b)

10.2 The first law of thermodynamics

The first law is just the law of conservation of energy applied to thermodynamics.

$$\underbrace{Q}_{\text{heat}} = \underbrace{\Delta U}_{\text{change in internal energy}} + \underbrace{W}_{\text{work done}}$$

- $Q > 0$: heat is supplied to the system; $Q < 0$: heat is removed from the system
- $\Delta U > 0$: internal energy / temperature increases; $\Delta U < 0$: internal energy / temperature decreases
- $W > 0$: work is done by the system (gas expands); $W < 0$: work is done on the system (gas compressed)

10.3 The second law of thermodynamics

克劳德·香农（Claude Shannon）说过：”熵是信息的度量。”永远的现代信息学之父，致敬。
It is quite extraordinary that this formula of thermodynamics has featured extensively in Claude Shannon's theory of information, a crucial part of modern telecommunications theory.

Entropy: a measure of the disorder of a system, or the number of ways the system can be arranged.
The change in entropy is given by

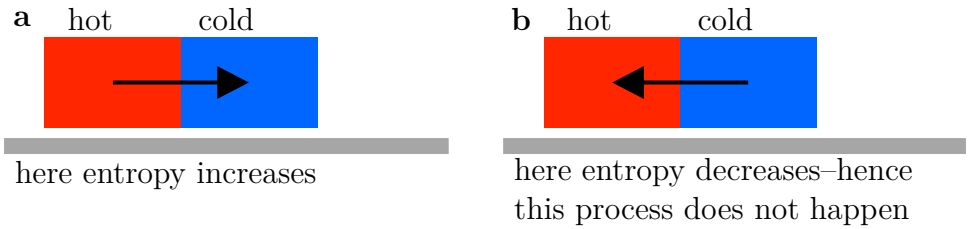
$$\Delta S = \frac{Q}{T}$$

where Q is the quantity of heat given to or removed from the system at a temperature T (in K).

- When heat is supplied to the system, $Q > 0$, $\Delta S > 0$, the entropy of the system increases.
- When heat is removed from the system, $Q < 0$, $\Delta S < 0$, the entropy of the system decreases.

For a reversible process that returns the system to its original state, the total change in entropy is zero (an example is the isothermal expansion / compression of gas).
Extension: $\Delta S = \int \frac{dQ}{T}$ when the temperature is not constant.

The entropy of an isolated system never decreases. Entropy increases in realistic irreversible processes and stays the same in theoretical, idealized, reversible processes.



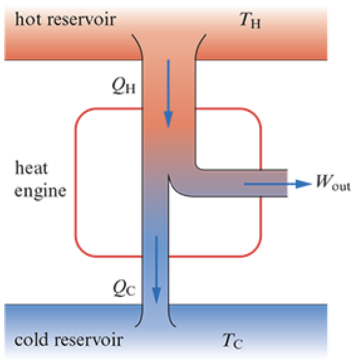
We denote $\Omega(H)$ the number of microstates that correspond to a macrostate H. (for example, the possible arrangements of N coins consisting of H heads)
Entropy is defined as:

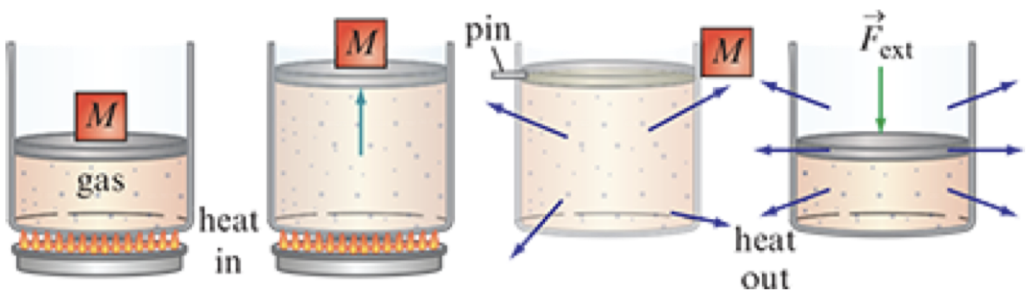
$$S = k_B \ln \Omega$$

where Ω is the number of microstates that correspond to the macrostate (e.g. total energy of the gas) of the system.
We see that the states which are likely to occur have large entropy.

10.4 Heat engines

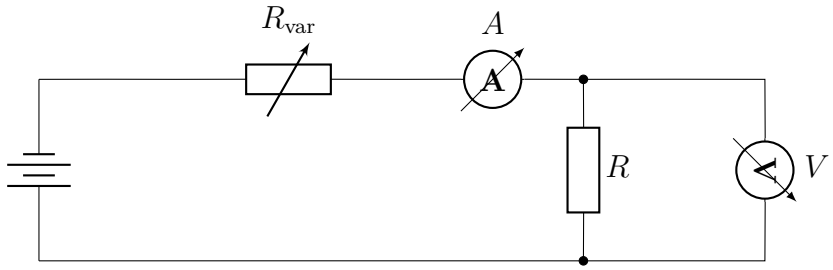
Definition: a device that transfers thermal energy to mechanical or electrical energy, which can then be used to do mechanical work.



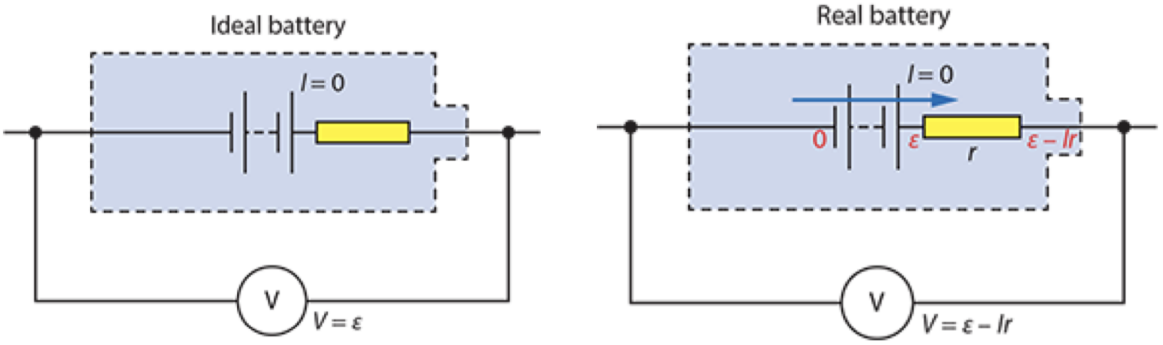


11.4 Terminal potential difference and the potential divider

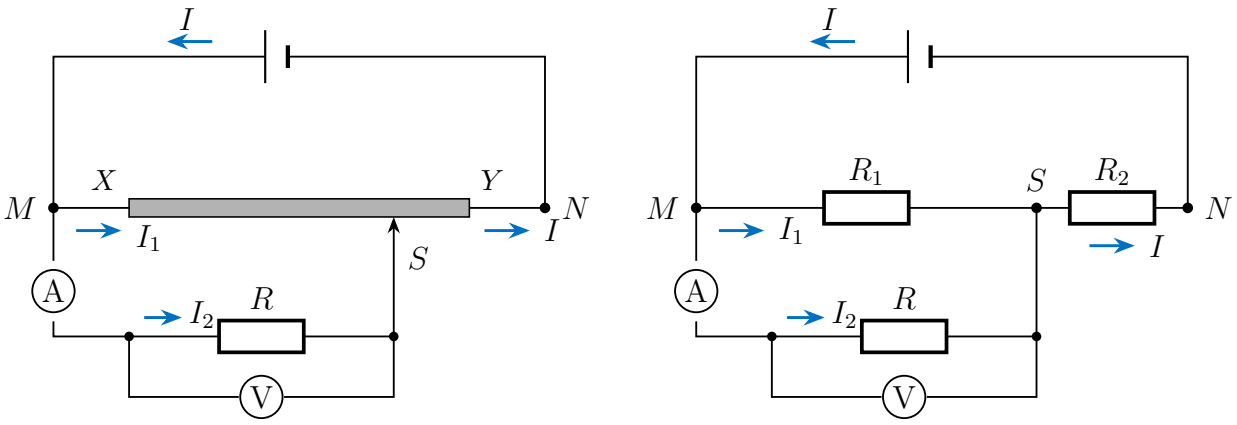
Ammeter: connected in series, after the component being measured, an ideal ammeter has 0 resistance.
Voltmeter: connected in parallel to the component being measured, an ideal voltmeter has infinite resistance.



The correct arrangement for measuring the current through and potential difference across a resistor.



We need to consider the existence of internal resistance in the real world: chemicals in cells (for example) also creates resistance in the cell.
The potential difference across the terminals of a battery is equal to the $e.m.f$ when there is no current in the circuit. It is less than the $e.m.f$ when there is current.
An $V - I$ graph can be used: ε is the y-intercept, r is the magnitude of the gradient of the line.



By varying where the slider S connects to XY, different potential differences and currents are obtained for the device R.
The potential difference across R now ranges from 0 to ε.

12.2 Details of simple harmonic motion

Displacement x as a function of time is expressed as

$$x = x_0 \sin(\omega t + \phi)$$

where x_0 is the amplitude and ϕ is the phase angle.

$$v = \omega x_0 \cos(\omega t + \phi);\ v_{\max} = \omega x_0$$

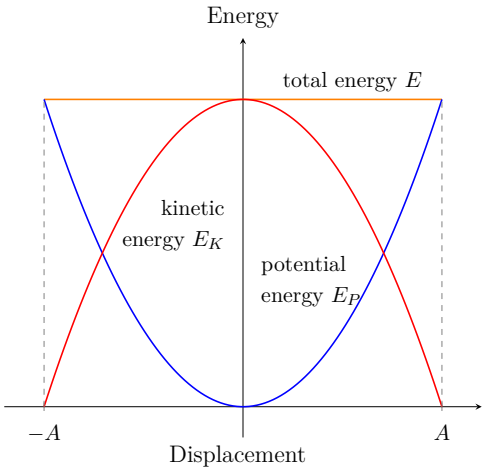
$$a = -\omega^2 x_0 \sin(\omega t + \phi);\ a_{\max} = \omega^2 x_0$$

Phase difference can be expressed as magnitude of differences between two phase angles.

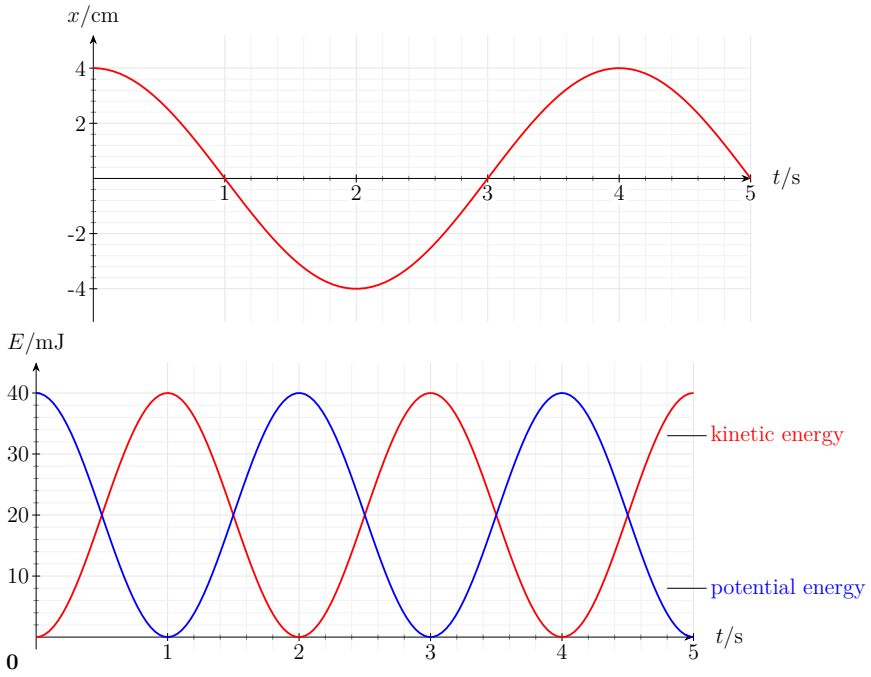
$$\Delta\phi = |\phi_1 - \phi_2|$$

12.3 Energy in simple harmonic motion

$E_T = E_P + E_K =$ constant (in the absence of other resistance forces, energy is conserved)



You will need to deduce how kinetic and potential energy varies with time from a given displacement-time graph:



12.4 More about energy in SHM

$$E_K = \frac{1}{2}m\omega^2(x_0^2 - x^2)$$

$$E_P = \frac{1}{2}m\omega^2 x^2$$

$$E_T = \frac{1}{2}m\omega^2 x_0^2$$

$$v = \pm\omega\sqrt{x_0^2 - x^2}$$

13 The wave model

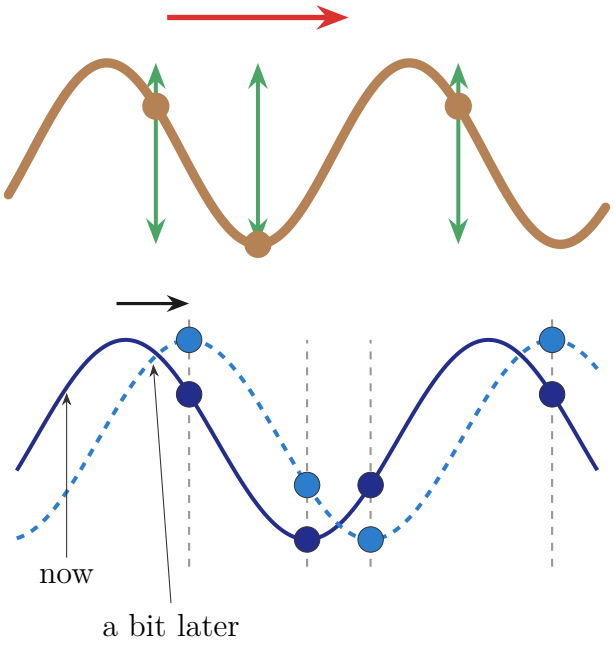
13.1 Mechanical pulses and waves

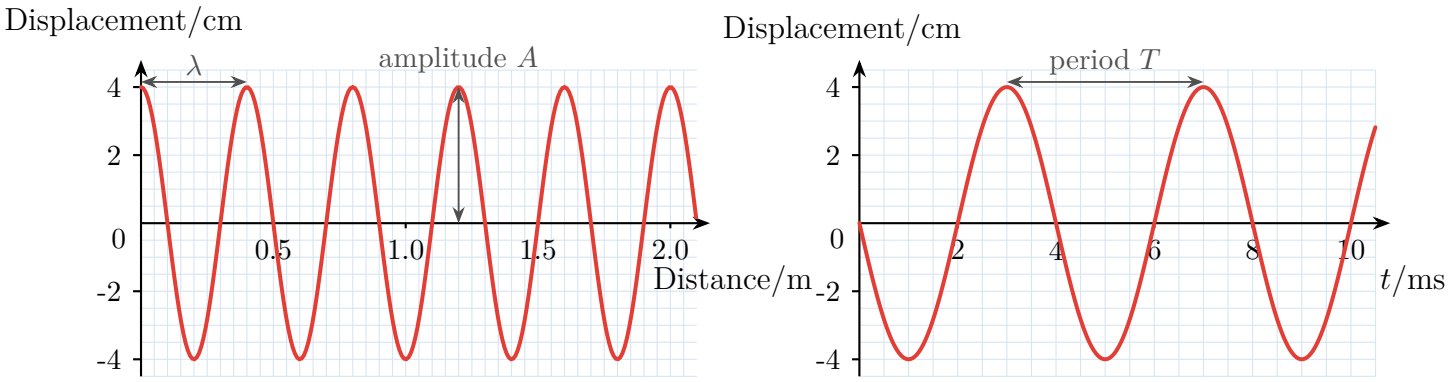
A mechanical wave is a disturbance that travels in a medium through oscillations of the particles of the medium. A wave transfers energy and momentum. The direction of propagation of the wave is the direction of energy transfer.

- Wavelength, λ, length of one full wave / distance from crest to crest or trough to trough
- Period, T, time to create one full wave
- Frequency, $f = \frac{1}{T}$, the number of full waves created in 1s
- Speed, $v = \frac{\lambda}{T} = f\lambda$

13.2 Transverse and longitudinal waves

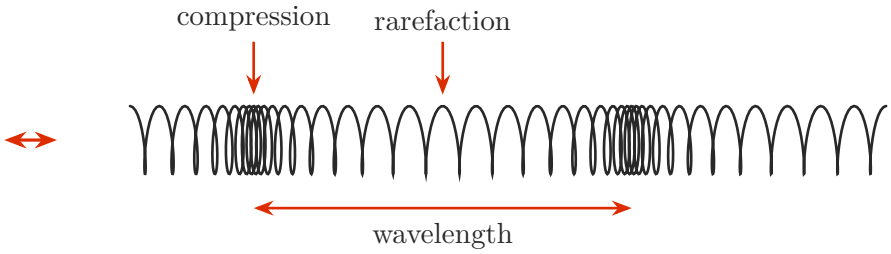
Transverse: the displacement of the particles of the medium is at right angles to the direction of energy transfer.
As a wave propagates through a medium, the particles of the medium perform SHM with a frequency that is the same as that of the wave.



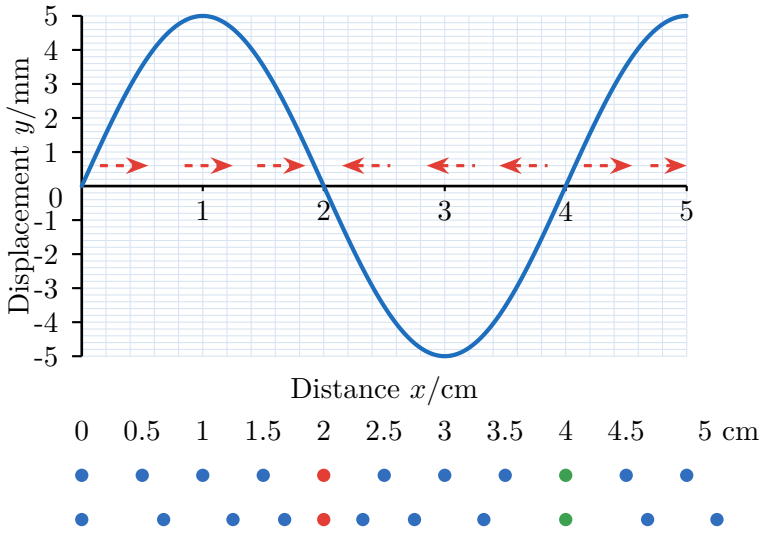


Displacement-distance and Displacement-time (displacement of one specific point)

Longitudinal waves: the displacement is parallel to the direction of energy transfer.



Compression: a region of higher than normal density
Rarefaction: a region of lower than normal density
You cannot tell whether a wave is transverse or longitudinal by looking at displacement-distance or displacement-time graphs. The graphs look the same for both.



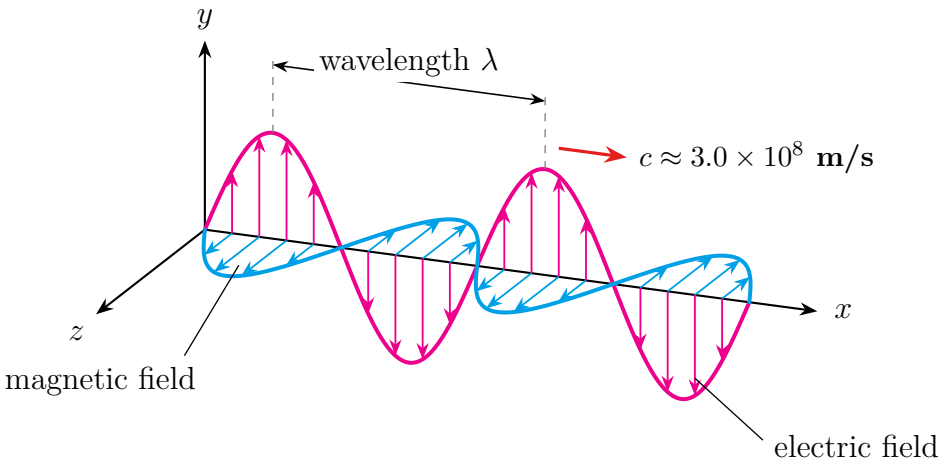
Positive displacement means moving right, negative displacement means moving left.

Phase difference is defined as

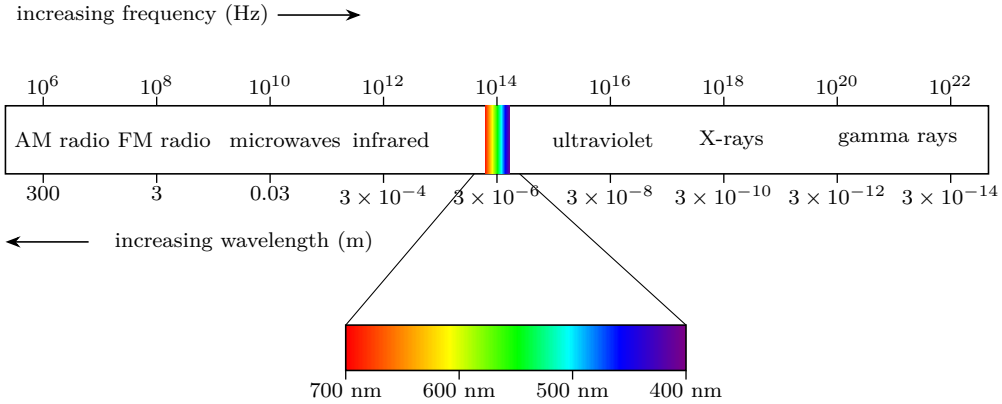
$$\Delta\phi = 2\pi\frac{\Delta x}{\lambda}$$

where Δx is the difference between consecutive peaks. The same formula can also be used to find the phase difference between two different points on the same wave.

13.3 Electromagnetic waves



An oscillating electric field produces an oscillating magnetic field such that the two are at right angles to each other and both propagate in space at the speed of light. Two fields are in phase.

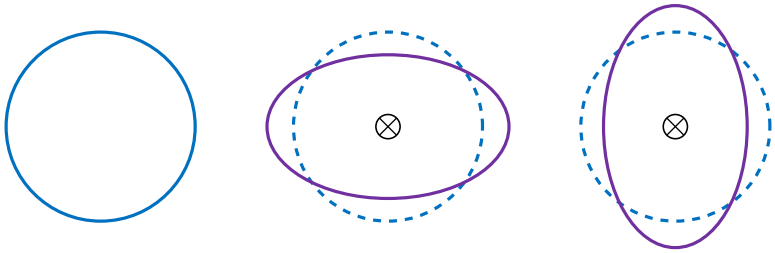


EM waves are transverse waves, and they can travel in a vacuum. They are produced when electric charges accelerate or during transitions in molecular, atomic or nuclear energy levels.

13.4 Waves extension

Gravitational waves

These waves would be generated whenever very large masses such as neutron stars and black holes collide with each other. Gravity waves are transverse and travel at the speed of light. The passage of a gravity wave through space distorts space, stretching and compressing it.



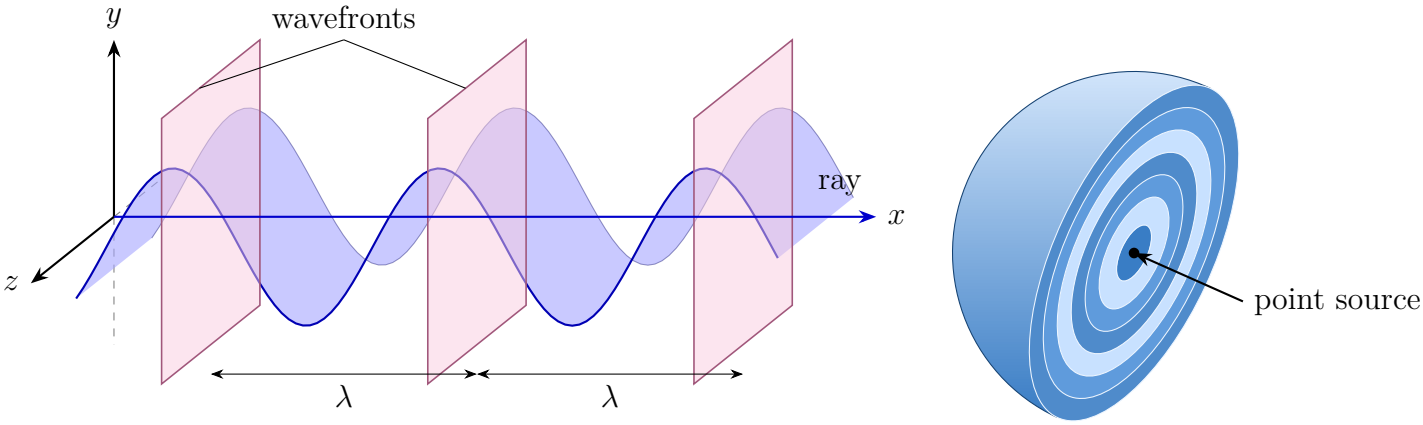
Matter waves

According to quantum theory, **all particles have wave properties**. The waves corresponding to particles are called matter waves. These are waves of probability: the displacement of the wave is related to the probability of finding a particle near a particular position.

14 Wave phenomena

14.1 Reflection and refraction

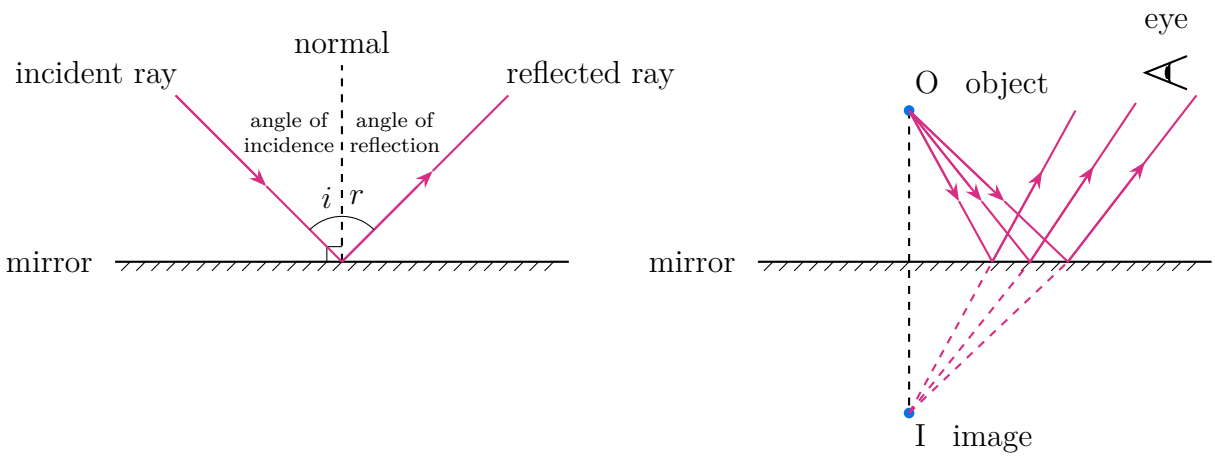
Wavefronts: planes will be normal to the direction of the wave.
Rays: lines at right angles to the wavefronts show the direction of wave propagation.



A point source emits waves in all directions. The wavefronts from a point source are spherical / cylindrical.

Reflection

Law of reflection: angle of incidence = angle of reflection, $i = r$.
Reflection takes place when the reflecting surface is sufficiently smooth. This means that the wavelength of the incident wave has to be larger than the size of any irregularities of the surface.
The wavelength of the reflected waves is the same as that of the incident wave.



Refraction

Refraction: the passage of light or any other wave from one medium into another where it has a different speed. It changes the direction of the incident ray. The wavelength changes for the wave in the medium. When a ray strikes an interface between two media, there is both reflection and refraction.

- If the speed of wave in medium is lower, the ray bends towards the normal.
- If the speed of wave in medium is higher, the ray bends away from the normal.

Snell's Law

$$\frac{\sin\theta_1}{c_1} = \frac{\sin\theta_2}{c_2} \Rightarrow n_1 \sin\theta_1 = n_2 \sin\theta_2$$

where n is the respective refractive index (must > 1, $\frac{\text{speed in vacuum}}{\text{speed in medium}}$).

Total Internal Reflection, TIR

TIR occurs at the boundary between two media when all the incident ray in medium 1 is reflected back into medium 1. For TIR to occur, the incident material must be denser than the second material. For incident angle $> c$, no ray will be refracted and TIR would occur.

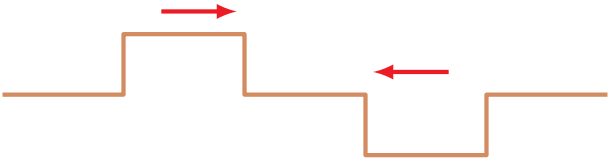
Applications of TIR include periscopes & **Optic fiber** (let's go CS)

When the angle of refraction is exactly 90° the light is refracted along the boundary. At this point, the angle of incidence is known as **the critical angle** c**.** Equation:

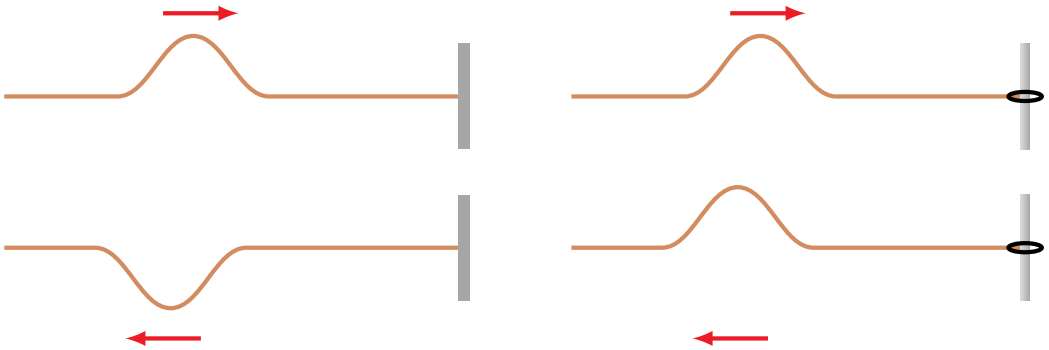
$$c = \sin^{-1}\left(\frac{1}{n}\right) \Leftrightarrow n = \frac{1}{\sin c}$$

14.2 The principle of superposition

想起了一些关于量子叠加态的内容..... $|k\rangle$
When two or more waves of the same type arrive at a given point in space at the same time, the displacement of the medium at that point is the algebraic sum of the individual displacements.



Reflection of waves



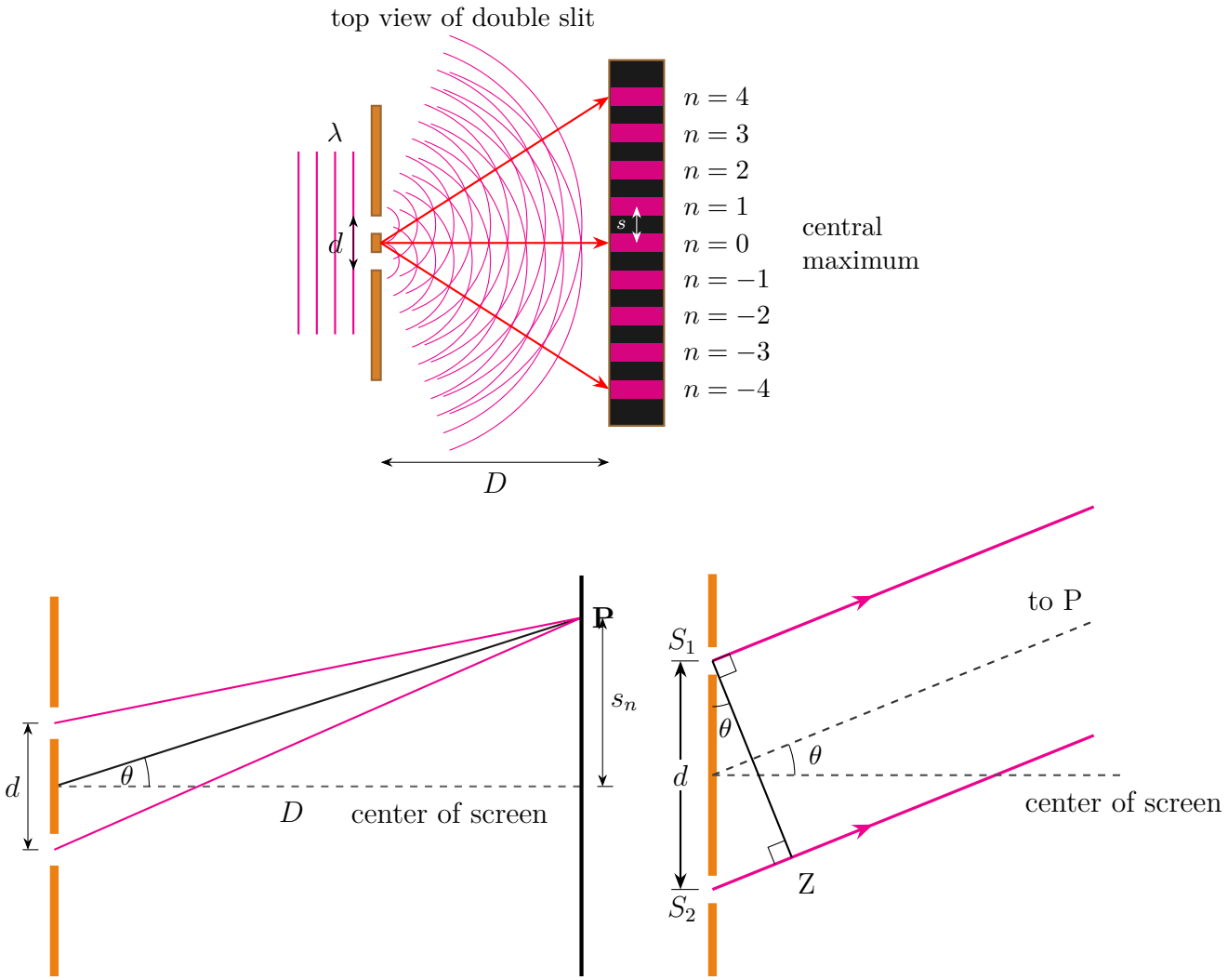
Fixed end, inverted (反作用力), phase change of 180°

End free to move, no phase change, end falls

14.3 Diffraction and interference

Diffraction: The spreading of a wave as it goes past an obstacle or through an aperture.
Diffraction is most prominent when the wavelength λ is larger than or equal to the gap width.

Young's Double Slit Experiment



The value of n indicates that the path difference is $n\lambda$.

Here, the path difference is $S_2Z = d\sin\theta$.
Deduction process for the linear separation on a screen of two consecutive maxima, s:

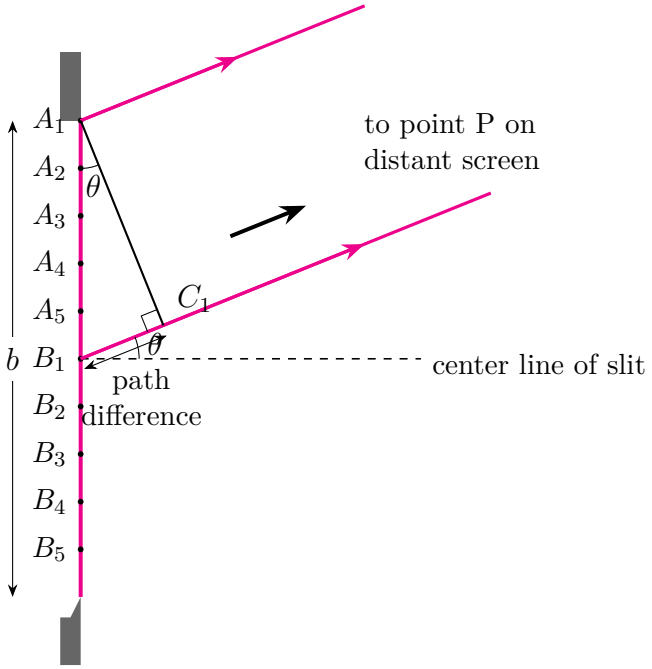
- Because d is small, $\sin\theta \approx \tan\theta \approx \theta$.
- From left diagram, $\tan\theta = \dfrac{s_n}{D} \Rightarrow s_n = D\sin\theta$
- $\sin\theta = \dfrac{n\lambda}{d} \Rightarrow s_n = \dfrac{n\lambda D}{d}$
- $s = \dfrac{\lambda D}{d}$, the maxima of the interference pattern are equally separated.

Coherent waves: phase difference between two waves (originated from the same source) does not change with time.

An intensity of one unit corresponds to the intensity from only one slit. At points of constructive interference the amplitude is double that of just one wave. Since intensity is proportional to the square of the amplitude, the intensity is four times as large.

14.4 Single-slit diffraction

Diffraction is negligible is λ is much smaller than the gap width and obvious when $\lambda \geq$ gap width.

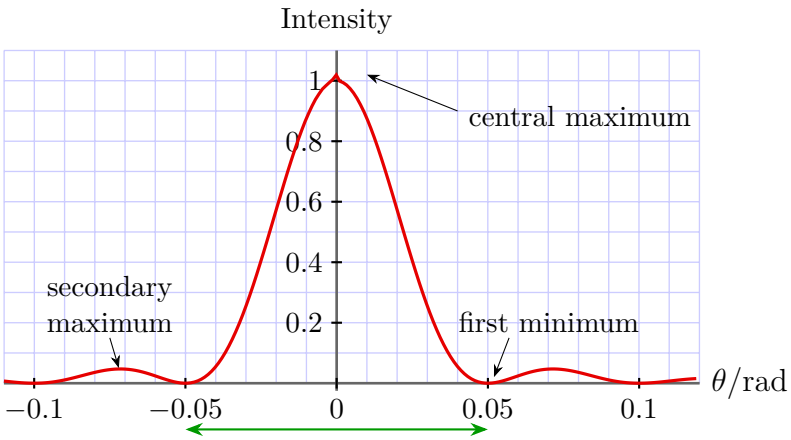


Points from A_1 to A_5 and from B_1 to B_5 are in phase (lies on the same wavefront).
The results of the interference of the waves can be seen on a distant screen.

The path difference between ray A_1 and B_1 is equal to $B_1C_1 = \frac{b}{2}\sin\theta$. This would result in complete destructive interference if the path difference is half of the wavelength. $\frac{b}{2}\sin\theta = \frac{\lambda}{2} \Rightarrow b\sin\theta = \lambda$

For every point on the upper half of the slit there is a corresponding point on the lower half, so we see that all the points on the wavefront will result in complete destructive interference if the first pair results in destructive interference.

Approximate formula for the angle of the first diffraction minimum: $\theta = \frac{\lambda}{b}$, where b is the slit width.



Intensity in single slit diffraction for red light.

The intensity of the first secondary maximum is only about 4.5% of the intensity at the central maximum.

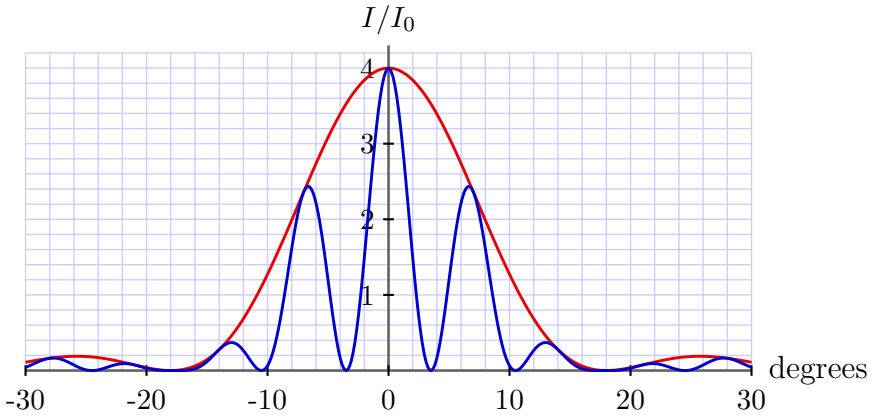
The angular width of the central maximum (measured by the green arrow) is given by $\frac{2\lambda}{b}$.

If we increase the slit width, more light goes through so the intensity will increase and the angular width will decrease.

Blue light has a shorter wavelength and so with blue light the first minimum occurs at smaller angle.

Effect of (non-negligible) slit width on two-slit intensity patterns

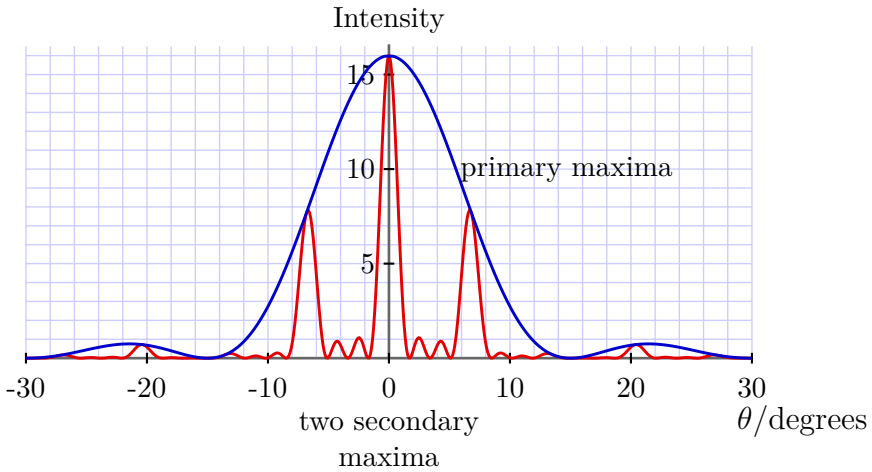
The single slit diffraction pattern modulates the two-slit pattern so that the intensity is given by the blue curve. The effect of the slit width is that the red curve becomes the envelope of the blue curve.



14.5 Multiple slits

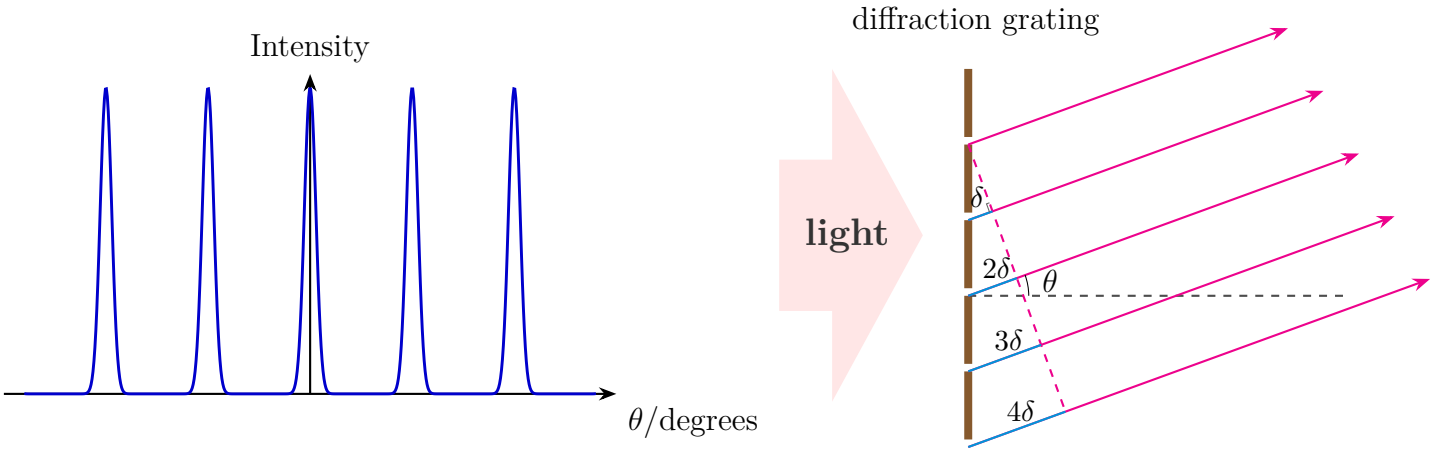
As the number of slits increases, the interference pattern increases in complexity. With N slits, there are $N-2$ secondary maxima. The intensity of the central maximum is N^2 times the intensity of just one slit by itself.
As the number of slits increases:

- The primary maxima are observed at the same angles
- The primary maxima get narrower and brighter
- The secondary maxima become unimportant



Diffraction Grating

Used to measure wavelength of light. A diffraction grating consists of a large number of parallel slits of very small width. (Instead of actual slits, modern gratings consist of a transparent slide on which rulings or grooves have been precisely cut.) The maxima in the interference pattern are sharp and bright and can easily be distinguished from their neighbours, they are called ”lines”.



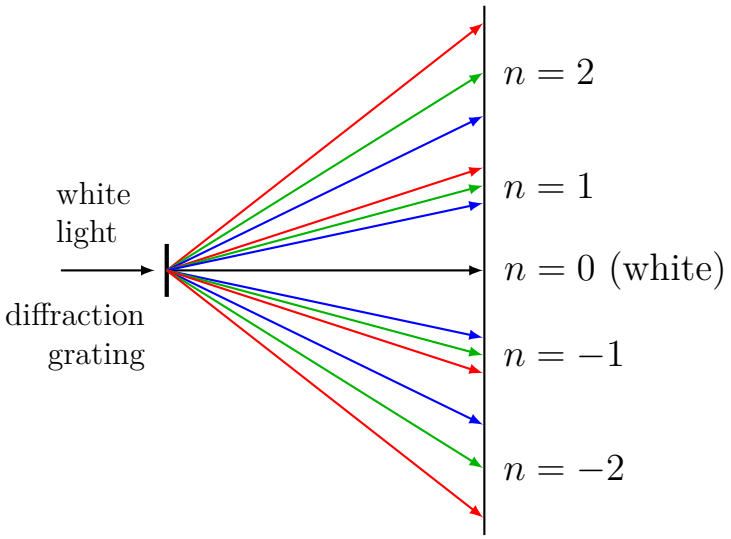
The intensity distribution of diffraction grating.
The maxima have roughly the same intensity and are very narrow.
右图中 $\delta = d\sin\theta$ 即为 path difference.

The maxima of the pattern are observed at angles that can be found by:

$$d\sin\theta = n\lambda, n \in \mathbb{Z}$$

A maximum observed at $n = k$ is called the kth order maximum.
In practice, "x lines per mm" $\Leftrightarrow d = \dfrac{1}{x}$ mm.

White light incidence on diffraction grating:



The $n = 1$ and $n = 2$ orders are beginning to overlap. This is because different wavelengths in different orders can satisfy the grating equation at the same angle, so light from the second order falls on top of light from the first order.

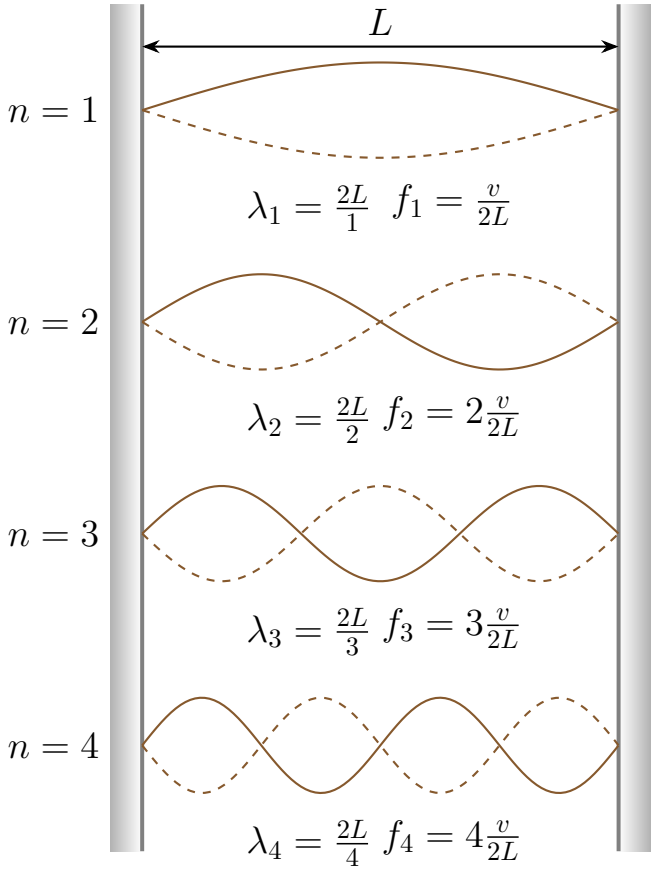
15 Standing waves and resonance

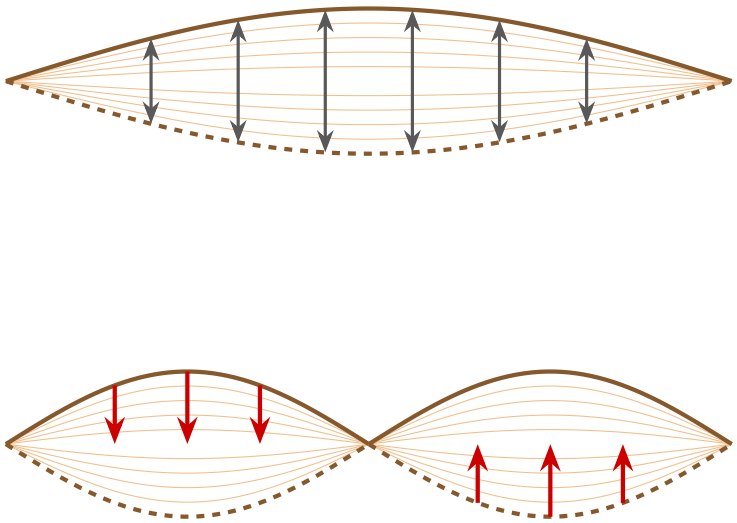
15.1 Standing waves

A standing wave is formed if two waves of the same speed, wavelength and amplitude travelling in opposite directions meet. The resulting wave has a displacement that is the sum of the displacements of the two travelling waves.
Characteristics of standing waves:

- The position of crests and troughs are always at the same position.
- Nodes: as a result of destructive interference, the displacement at that point is always zero. The distance between two consecutive nodes is half a wavelength.
- Antinodes: points halfway between two nodes, as a result of constructive interference, the displacement gets as large as possible (for an instant of time only).
- The amplitude of oscillation is different at different points on the string.
- A standing wave does not transfer energy.
- The ends of a standing wave are either nodes or antinodes. These "boundary" or "end conditions" determine the possible wavelength of the wave.
- All points on a standing wave have the same frequency.

15.2 Standing waves on strings





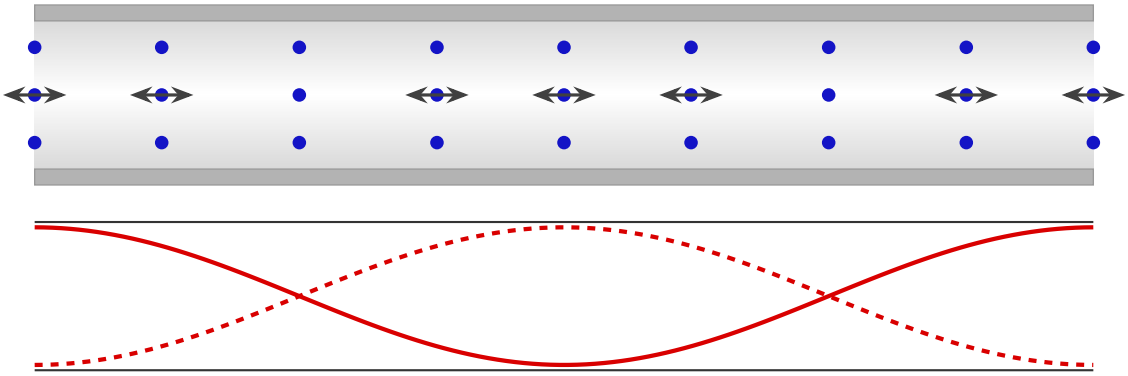
* : the λ is deduced using the fact that the distance between two consecutive nodes is half a wavelength.

Different standing wave patterns will be established on the string depending on the frequency used.
The standing wave that has the longest wavelength here is called the first harmonic.
The higher harmonics have more loops, and the wavelength decreases and the frequency increases.
Generally speaking, $\lambda_n = \dfrac{2L}{n}$. (while f is given by $\dfrac{v}{\lambda}$, where v is the speed of the travelling wave).

Different points on the standing wave oscillate with different amplitudes.
Points within the same loop oscillate in phase, while there is a 180° phase difference between points in one loop and points in the next loop.

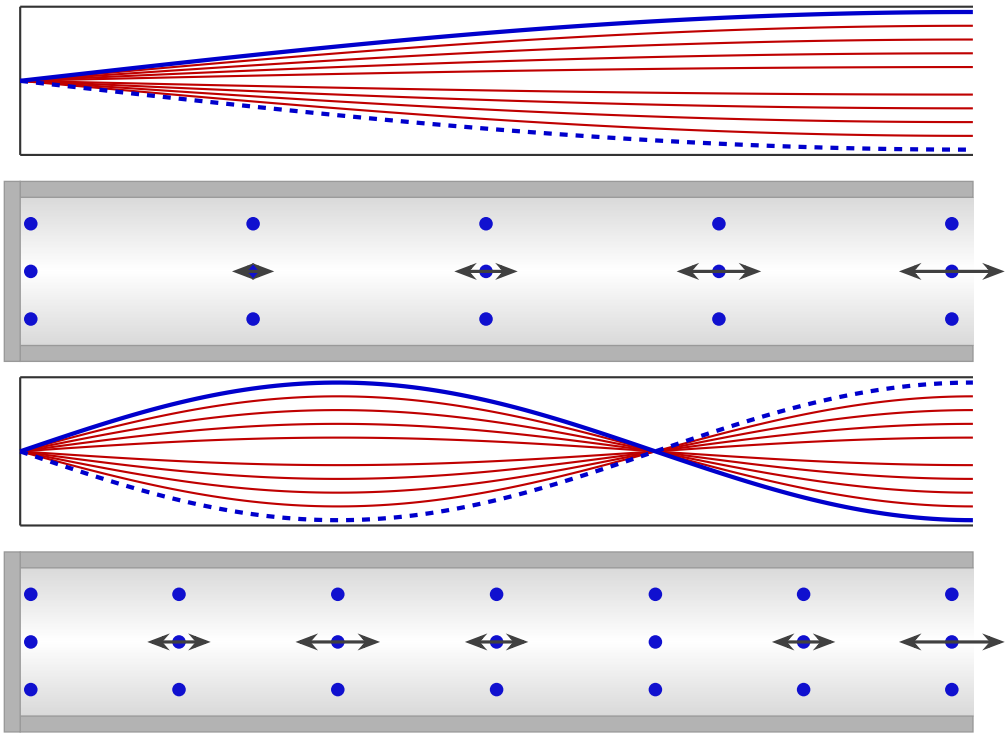
Note that when asked about sound made from the string, the f of the standing wave is the same as that of the sound.

15.3 Standing waves in pipes



The case of a pipe with both ends closed (which is not very useful) is similar to that of a string with ends fixed (two nodes at the ends). Normally, only discuss about the case of both end opening.

The dots represent molecules of air in the pipe. The double-headed arrows show how far these molecules oscillate back and forth. The molecules at the ends oscillate the most.



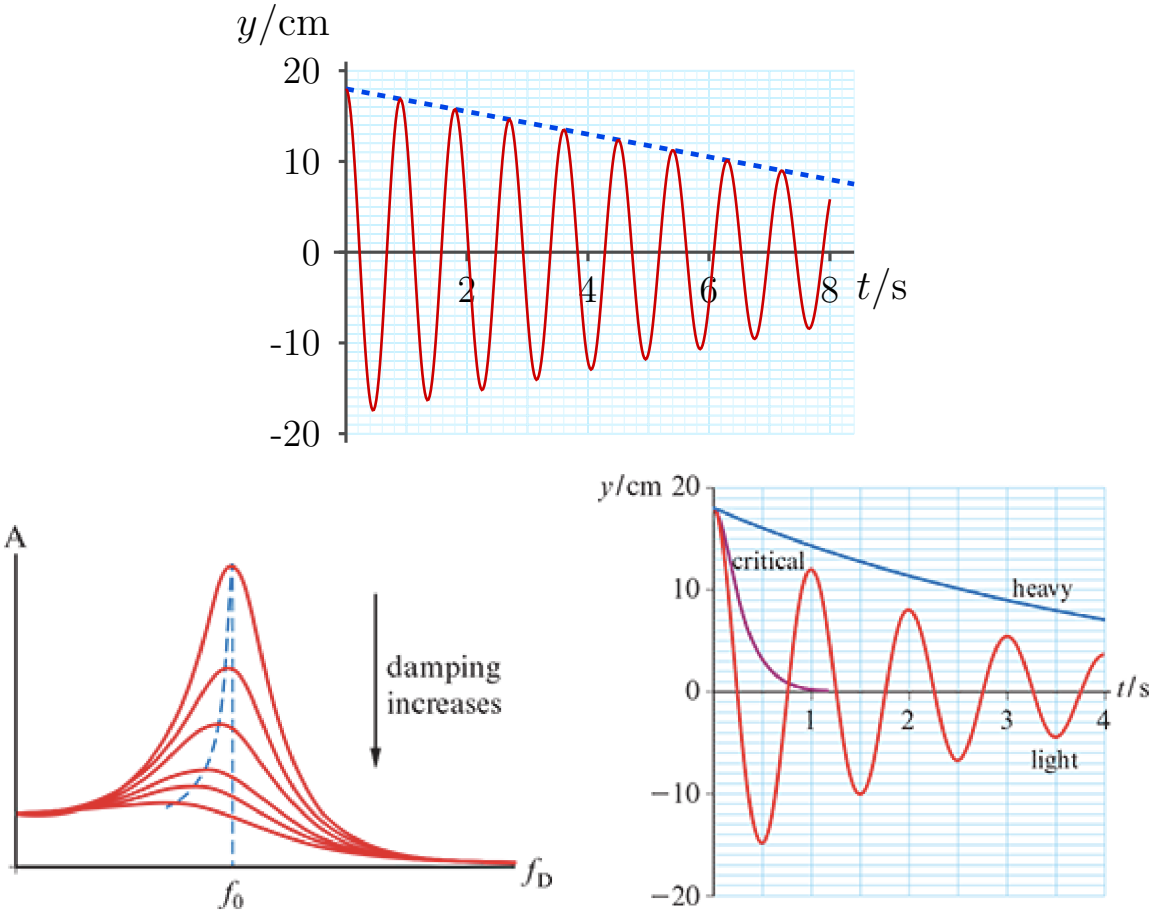
* : The case of a free end (antinode) and a closed end (node).

For this case, the allowed wavelength is given by $\lambda = \dfrac{4L}{n}$, n is an odd number.

15.4 Resonance and damping

If the externally applied forces resist the motion we will have damping, loss of energy in an oscillating system. The amplitude decreases exponentially with time.

Natural frequency f_0: the frequency that the system is free to oscillate.
The force F will be assumed to vary periodically with time with the driving frequency, f_D.
Oscillations that happen under externally applied force F is said to be forced / driven.
After the external force is first applied, the system will eventually switch to oscillations with a frequency equal to f_D. This is true both when the system is initially at rest and when it is oscillating at its own frequency before the force is applied.
The amplitude of the oscillations will depend on f_D and the amount of damping.



The degree of damping increases as we move from the top curve down.
The blue dashed curve represents the peak of the curves (maximum amplitudes).

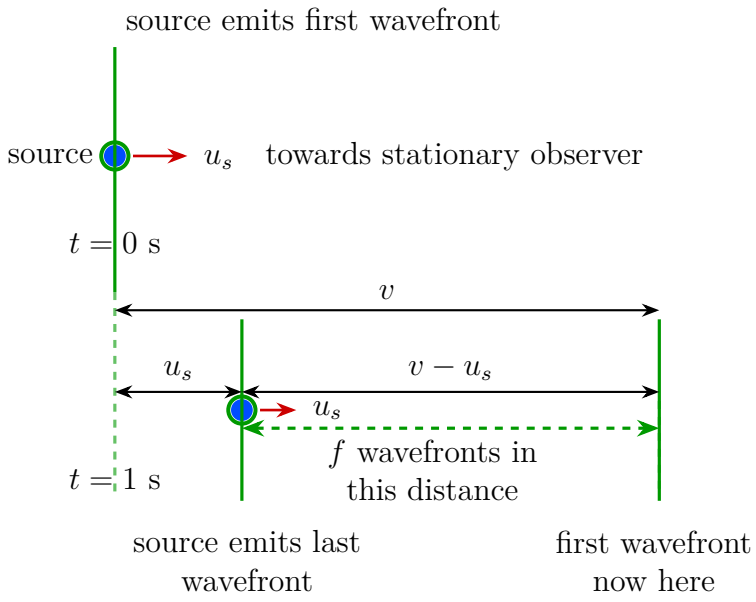
- For very small damping, the peak of the curve occurs very close to the natural frequency of the system. When the driving frequency is equal to the natural frequency, the amplitude of oscillations is a maximum. This condition is called **Resonance**.
- As the amount of damping increases, the peak shifts to lower frequencies and becomes wider and shorter. In other words, with large damping the amplitude is a maximum when the driving frequency is slightly less than the natural frequency.
- At very low frequencies, the amplitude is essentially constant, and at high frequencies the amplitude goes to zero.

Three cases regarding damping:

- In light damping the system returns to equilibrium after many oscillations of decreasing amplitude.
- In critical damping, the system returns to equilibrium in the shortest possible time without performing oscillations.
- In the heavy or overdamped case the system returns to equilibrium after a very long time without performing oscillations.

16.2 The Doppler effect for sound

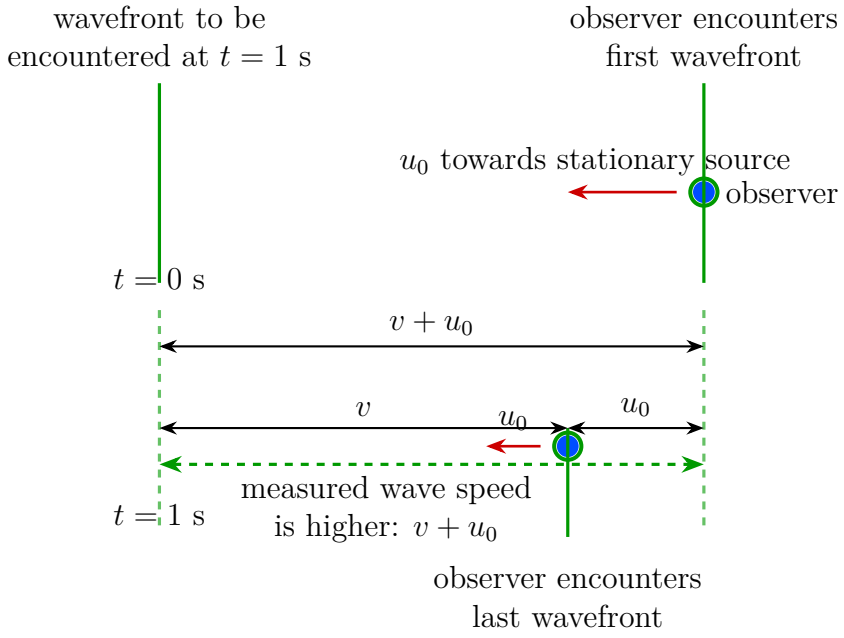
Moving source and stationary observer



The stationary observer will measure a wavelength of $\lambda' = \frac{v - u_s}{f} = \frac{v}{f'} = \lambda\left(1 \pm \frac{u_s}{v}\right)$ (where + is for source moving away, - for source approaching).

The frequency measured would then be $f' = \frac{v}{\lambda'} = f\left(\frac{v}{v \pm u_s}\right)$ (where + is for source moving away, - for source approaching).

Stationary source and moving observer



It could be seen as, from the observer's perspective, that the source is moving. Note that here, the speed of sound relative to the observer is no longer v as the medium has turned from still air to moving air.

Frequency observed would be given by $f' = f\frac{v + u_0}{v + u_0 - u_0} = f\left(\frac{v \pm u_0}{v}\right)$ (where + is for observer approaching, - for observer moving away) The wavelength would be the same.

Unit D Fields

17 Gravitation

17.1 Newton's law of gravitation

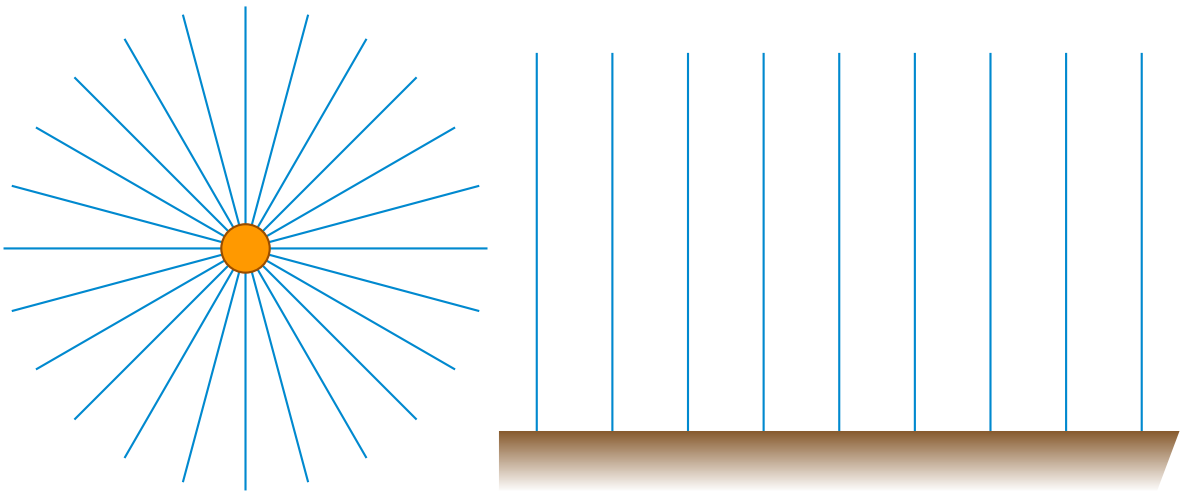
A planet that orbits around the sun experiences acceleration, and thus a force is acting on it.
The attractive force of gravitation between two point masses (idealised bodies with no physical size) are:

$$F = G\frac{M_1 M_2}{r^2}$$

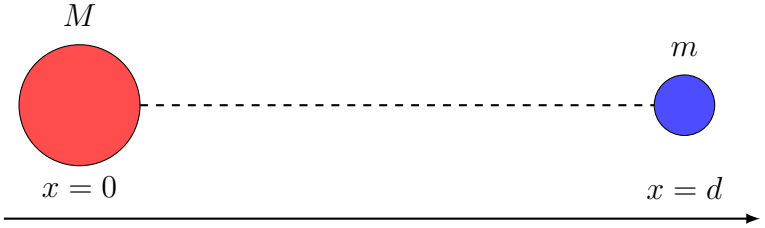
where $G = 6.67 \times 10^{-11}\ \mathrm{N\,m^2\,kg^{-2}}$, and r is the center-to-center distance between the spherical bodies.
The direction of the force is along the line joining the two masses.
The gravitational force is always attractive. The magnitude of the force on each mass is the same.
(Spherical bodies with uniform distribution of mass can be considered as point mass, as the center of mass lies within the center of the spherical body.)

Gravitational Field Strength

A mass M is said to create a gravitational field in the space around it. A formal definition: region of space where a mass experiences a (gravitational) force.
The gravitational field strength, at a certain point, is the gravitational force per unit mass experienced by a small point mass m placed at that point: $g = \frac{F}{m}$, or $\frac{GM}{r^2}$ for spherical bodies with mass M.

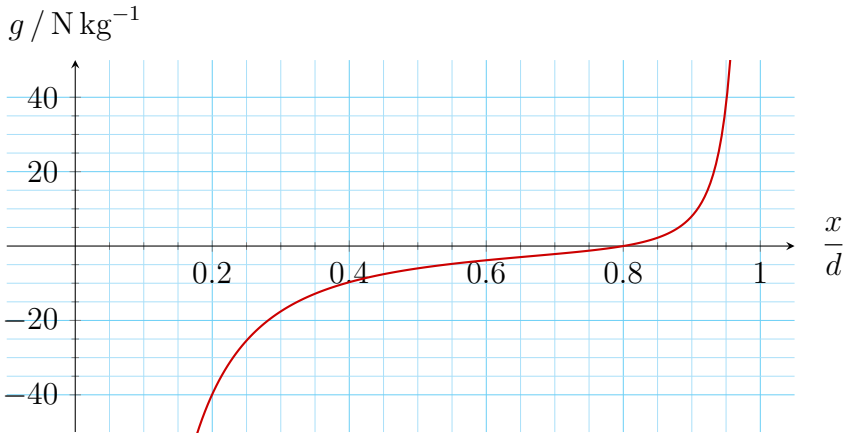


The gravitational field strength around a single point or spherical mass is radial, which means that it points towards the centre of the mass creating the field. The field is not uniform: magnitude of g decreases as we move away from M.
The other case: a uniform field with constant gravitational field strength. (e.g. constant a of freefall)



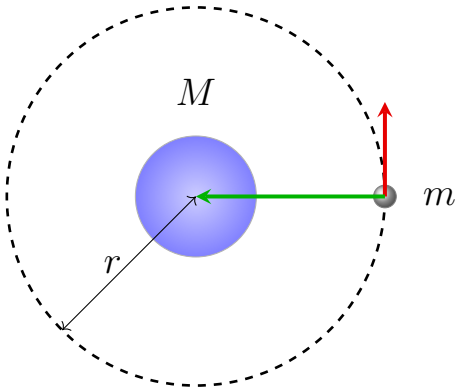
At a point that is a distance x from the centre of the larger mass the net gravitational field strength is:

$$g = -\frac{GM}{x^2} + \frac{Gm}{(d-x)^2}$$



Variation of g with distance from the larger mass.

Orbital Motion

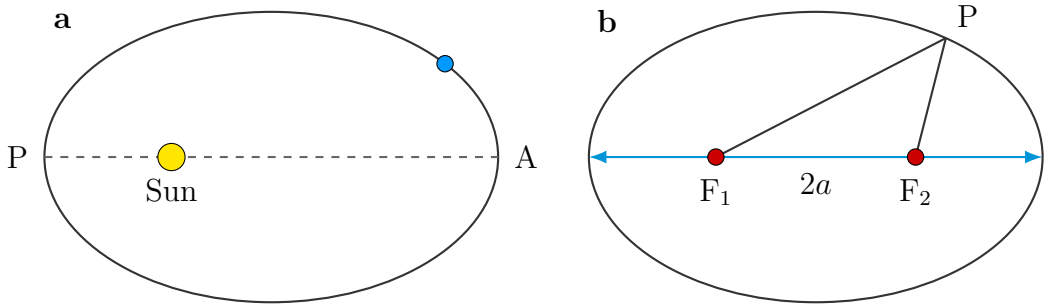


$$v = \sqrt{\frac{GM}{r}} \; ; \; T^2 = \frac{4\pi^2 r^3}{GM}$$

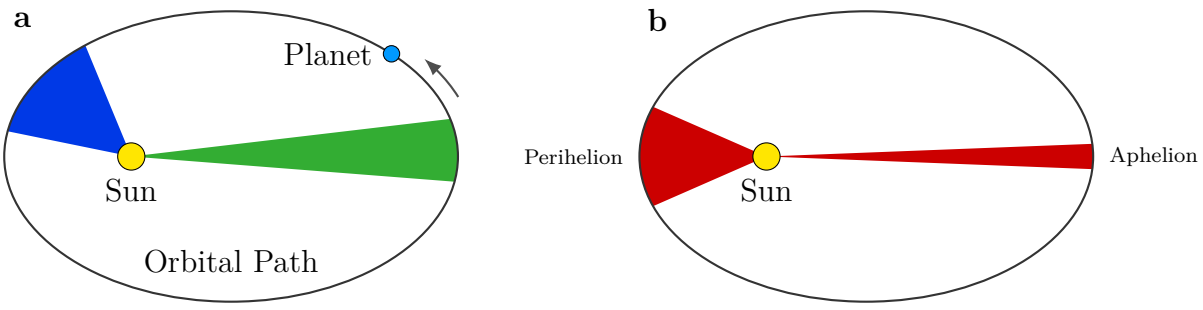
where T is the time for one complete revolution, period of revolution.

Kepler's Laws

- First Law: planets move on ellipses with the sun at one of the foci of the ellipse.



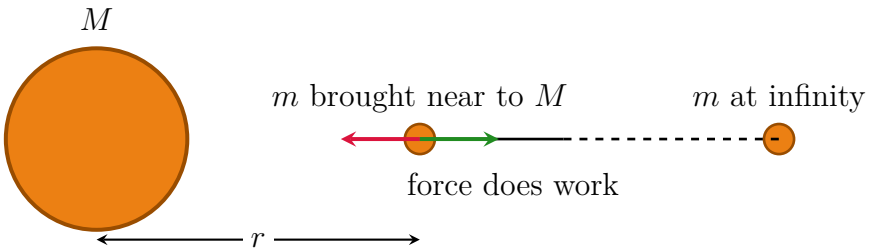
- Second Law: the line joining a planet and the sun sweeps equal areas in equal times.



- Third Law: the period of revolution of a planet around the sun is proportional to the $\frac{3}{2}$ power of the semi-major axis of the ellipse. $\left(T^2 = \frac{4\pi^2 a^3}{GM}\right)$.
  For a circular orbit with radius r, $r = a$.

17.2 Gravitational potential and energy

The gravitational potential energy of two bodies is the work that was done in bringing the bodies to their present position from when they were infinitely far apart. This energy is stored in the gravitational field.



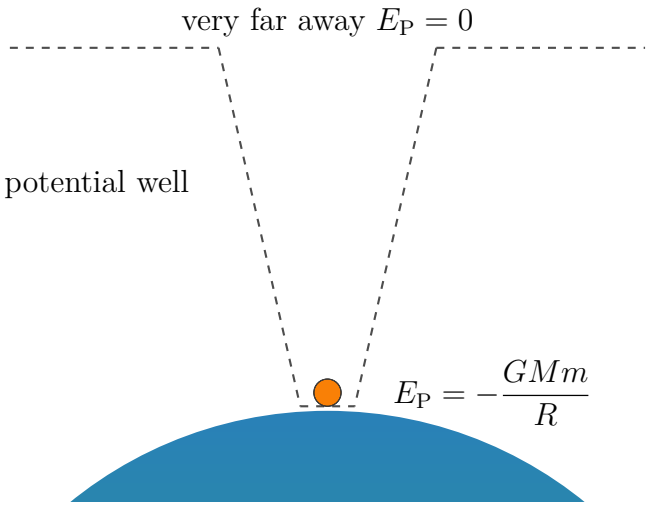
* : Assume M is fixed and that only m has moved.
This energy belongs to the pair of masses M and m and not just to one of them.
(Assume no kinetic energy is involved. Masses move at a small constant speed.)

The total work done in moving the mass m from infinity to a distance r from the centre of a spherical mass M is:

$$W = \int_{\infty}^{r} \frac{GMm}{r^2} \mathrm{d}r = -\left[\frac{GMm}{r}\right]_{\infty}^{r} = -\frac{GMm}{r} \Rightarrow E_p = -\frac{GMm}{r}$$

Why is the gravitational potential energy negative:

- Note that the force is directed to the right, while the motion is directed left.
- No potential energy initially $\Rightarrow$ Increase E_K for m and decreased potential energy overall



Gravitational Potential

The gravitational potential, V_g, at a point P in a gravitational field is the work done per unit mass in bringing a small point mass from infinity to point P.

$$V_g = \frac{W}{m} = -\frac{GM}{r};\ E_p = mV_g;\ W = m\Delta V_g$$

where W is the work done on the mass by the external agent moving the mass.

Field and potential

Gravitional field strength can be calculated using:

$$g = \frac{\Delta V_g}{\Delta r} \left(= -\frac{dV_g}{dr} \right)$$

where Δr is the difference between two points close to each other and ΔV_g is the difference between gravitational potential.

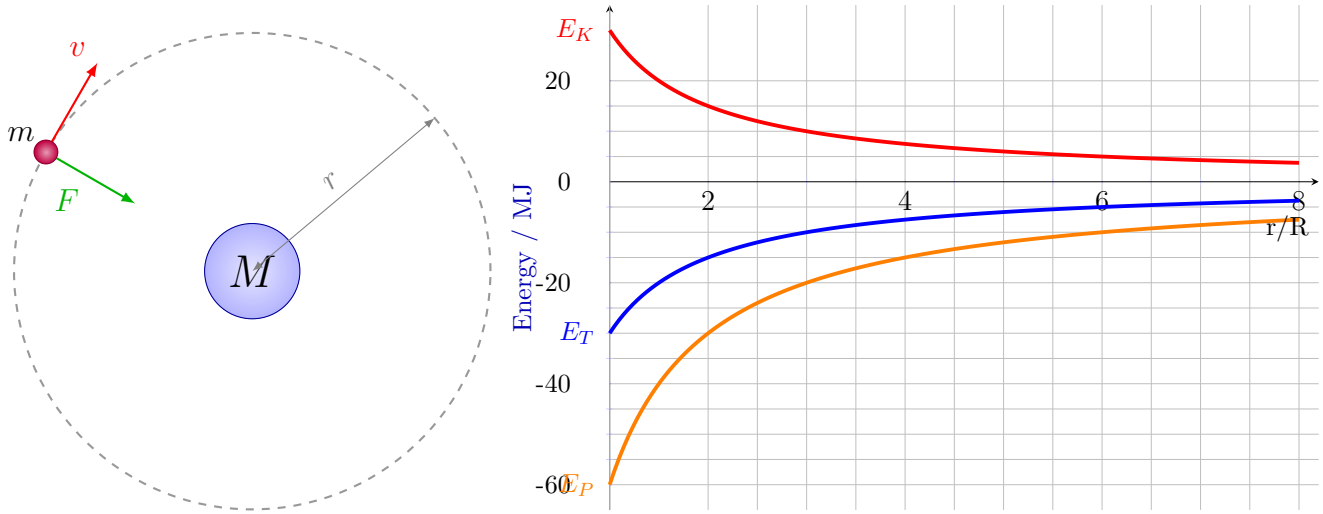
Equipotential surface

An equipotential surface consists of points (that have same distance from the source) that have the same gravitational potential.



Equipotential surfaces for two equal spherical masses and two unequal masses. Field lines are normal to the equipotential surfaces. The work done to move a point mass from one point to another point on the same equipotential surface (from A to B) is zero: $\Delta V_g = 0$.

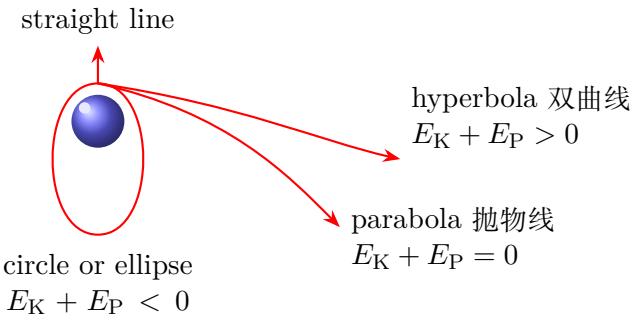
17.3 Motion in a gravitational field



$$E_T = E_K + E_p = \frac{1}{2}mv^2 - \frac{GMm}{r} = \frac{1}{2}m\left(\sqrt{\frac{GM}{r}}\right)^2 - \frac{GMm}{r} = -\frac{GMm}{2r}$$

- If the total energy is positive, the object will follow a hyperbolic path and never return.
- If the total energy is zero, the object will follow a parabolic path to infinity, where it will just about stop. It will never return.

- If the total energy is negative, the object will go into a circular or elliptical orbit (or crash into the planet if the launching speed is too low).



Escape speed

Escape speed is the minimum speed a body in a gravitational field must have in order to reach infinity.
The condition to escape from any point in a gravitational field is that the total energy at that point must be zero.

$$v_{esc} = \sqrt{\frac{2GM}{r}}$$

Reasons behind weightlessness

Essentially, the resultant force on the astronaut is zero.
In the case where W is the weight, F is the net force, N is the reaction force from the floor of the spacecraft:

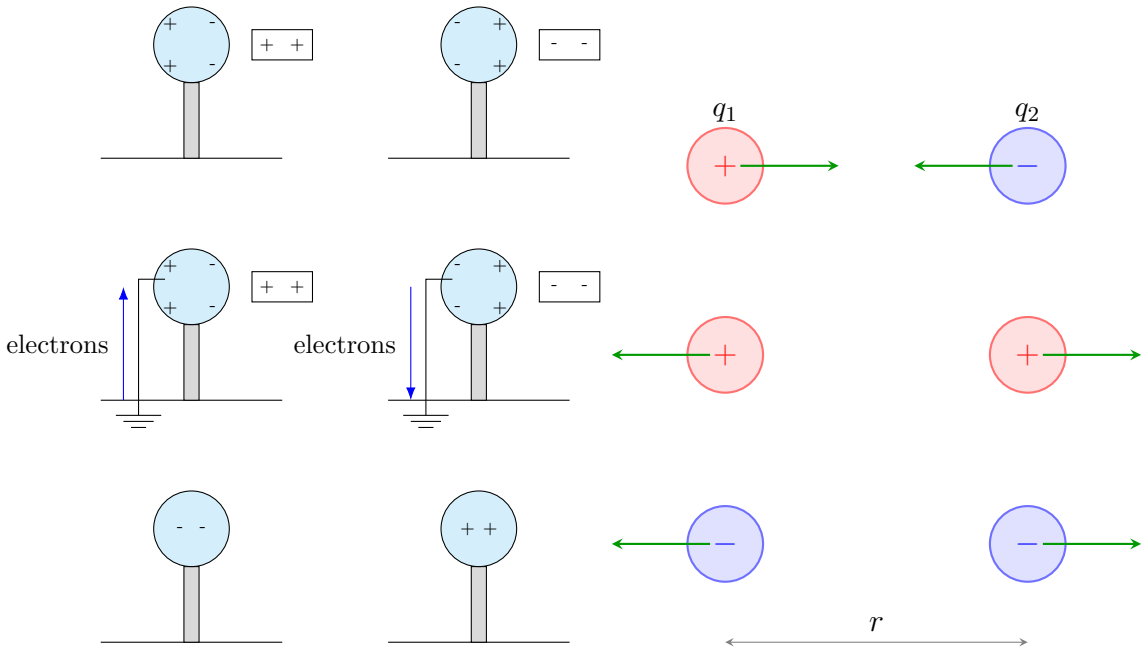
$$\begin{aligned} F &= W - N \\ &\Downarrow \\ N &= W - F \\ &= \frac{GMm}{r^2} - \frac{mv^2}{r} \\ &= \frac{m}{r}\left(\frac{GM}{r} - v^2\right) \\ \because v_{orbit}^2 &= \frac{GM}{r} \\ \therefore N &= 0 \end{aligned}$$

18 Electric and magnetic fields

18.1 Electric charge, force and field

Properties of electric charges:

1. Law of conservation applies to electric charges: they can't be created or destroyed.
2. It is a quantizied quantity: this means that the amount of electric charge on a body is always an integral multiple of a basic unit. ($1e = 1.6 \times 10^{-19}$ C)



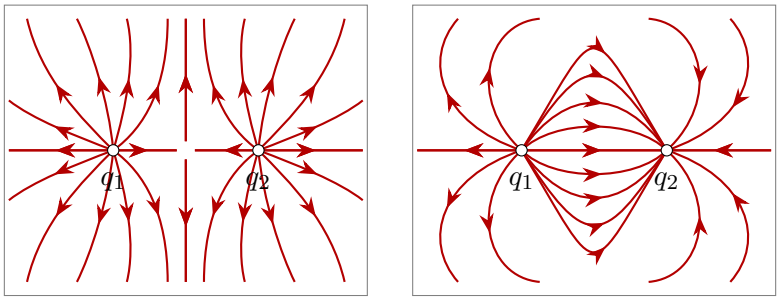
Coulomb's Law states that, the electric force between two point charges q_1 and q_2 that is separated by a distance of r is given by

$$F = k\frac{q_1 q_2}{r^2}$$

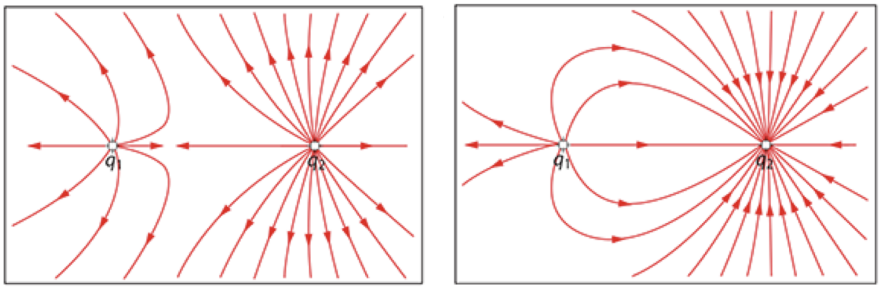
where $k = \dfrac{1}{4\pi\varepsilon}$ is the Coulomb constant (the ε is the permittivity).
($k = 8.99 \times 10^9$ N m^2 C^{-2}, $\varepsilon = 8.85 \times 10^{-12}$ C^2 N^{-1} m^{-2} (in vacuum))

Electric Field

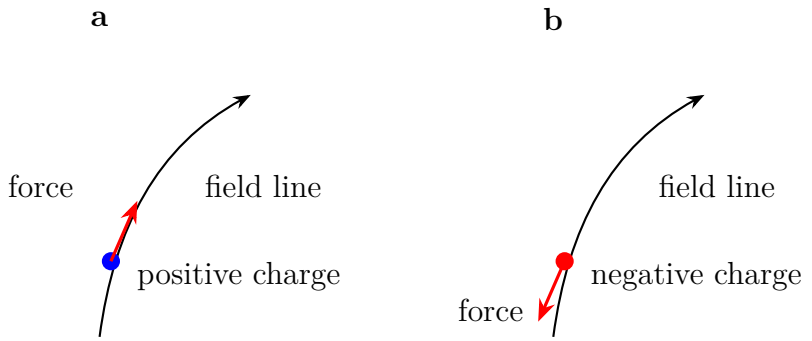
Electric Field Strength, E, is the electric force per unit charge experienced by a small, positive, point charge q: $\vec{E} = \dfrac{\vec{F}}{q}$. The direction of the electric field is the same as the direction of the force experienced by a positive charge at the given point.
The formula $E = k\dfrac{Q}{r^2}$ applies to a point charge / outside of the spherical charge.



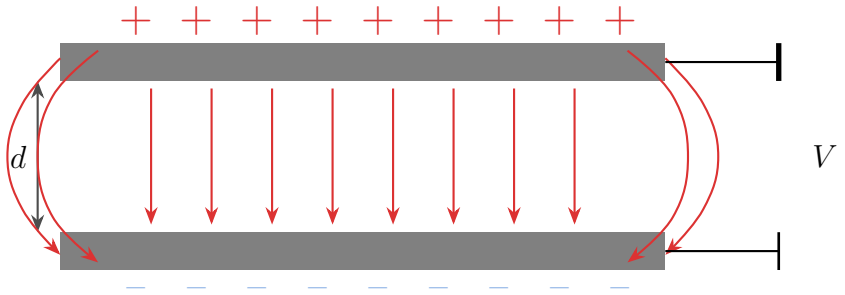
Field lines for two equal positive charges and two equal and opposite charges.



Field lines for unequal and positive charges. Field lines for unequal and opposite charges.

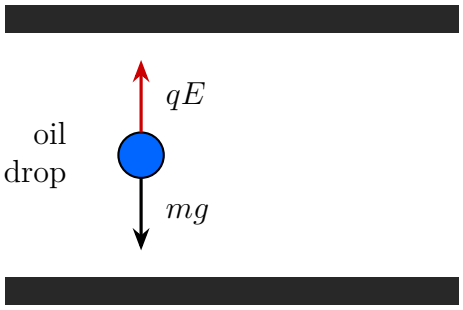


- The electric force on a positive charge is tangent to the field line in the direction of the arrow of the field line.
- The force on a negative charge is tangent to the field line in a direction opposite to the arrow.



The magnitude of the **uniform** electric field strength in between the plates is given by $E = \frac{V}{d}$.

The Millikan oil drop experiment

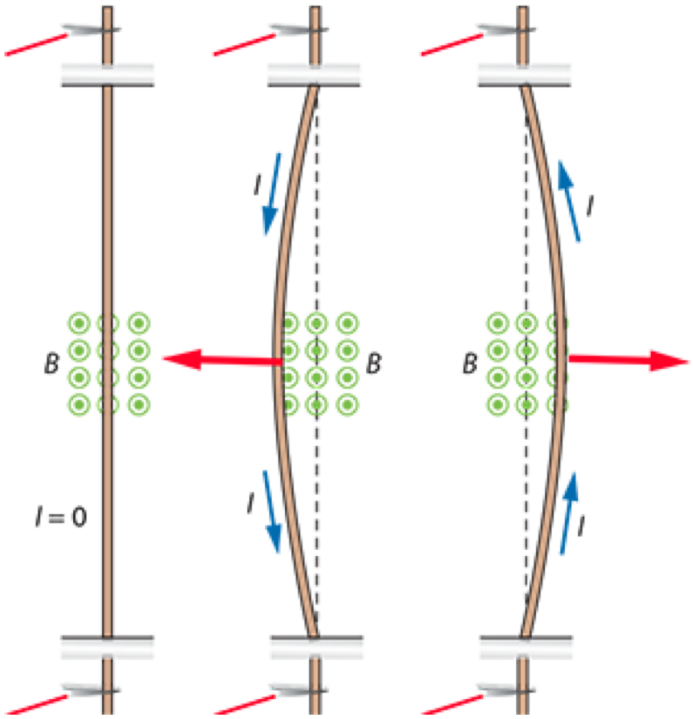


In this experiment, oil drops that were charged by X-rays were allowed to enter a region of uniform electric field between the parallel plates. Some oil drops were observed to be at rest in between the plates. This means that the weight of the oil drop is matched by the electric force.

$$mg = qE$$
$$mg = q\frac{V}{d}$$
$$\Downarrow$$
$$q = \frac{mgd}{V}$$

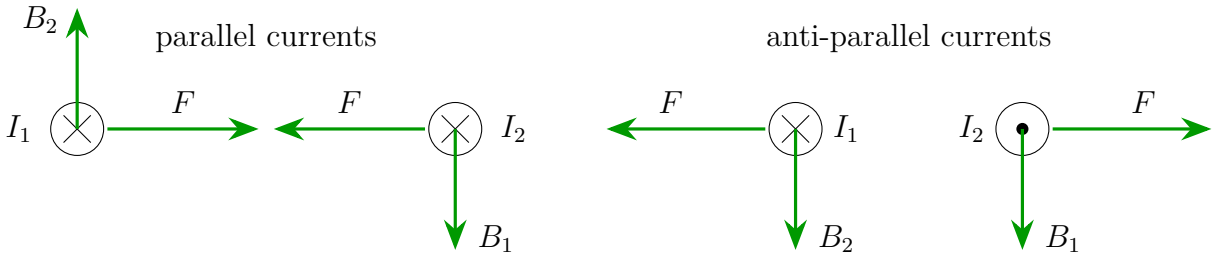
Millikan measured the mass of the oil drop by turning off the electric field and observed the oil drop that is now falling vertically down. The oil drop quickly achieves terminal speed because now the weight is matched by the drag force given by Stoke's Law:

where θ is the angle between the current and the direction of the magnetic field.



The force between two current-carrying wires

The currents are different in the two wires, so the magnetic fields are different, but the two forces are equal in magnitude. (In the diagram below, $B_1 \neq B_2$.) If the currents are parallel, the forces are attractive, and if they are anti-parallel, the forces are repulsive.



Extension: Magnitude of Magnetic Flux density created

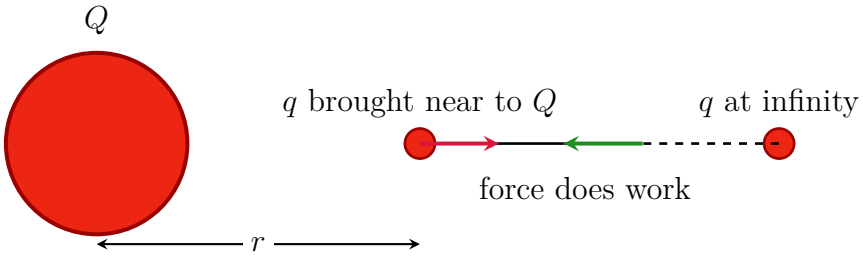
The magnitude of the magnetic flux density created at a distance r from a long straight wire carrying current I is given by the formula $B = \dfrac{\mu_0 I}{2\pi r}$, where $\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1} is a constant of physics called magnetic permeability.

The magnetic flux density created by the left wire at the position of the right wire is $B_1 = \dfrac{\mu_0 I_1}{2\pi r}$. Therefore the force experienced by a length L of the right wire is $F = B_1 I_2 L = \dfrac{\mu_0 I_1 I_2 L}{2\pi r}$.

Thus, we can obtain the formula for force per unit length: $\dfrac{F}{L} = \mu_0 \dfrac{I_1 I_2}{2\pi r}$.

18.3 Electric potential and electric potential energy

An electric charge creates electric potential. Two electric charges also share electric potential energy. The formulas we derived for gravitation carry over to electricity essentially by replacing mass everywhere by charge.



The electric potential at a point P in an electric field is the amount of work done per unit charge as a small positive test charge q is moved from infinity to P: $V_e = \frac{W}{q}$.
The work done in moving a charge q from infinity to point P is the electric potential energy of the system, $E_P = qV_e \left(= \frac{kQq}{r}\right)$. (Work done is expressed as $W = \frac{kQq}{r}$ by using calculus, therefore $V_e = \frac{kQ}{r}$. The value of $k = 8.99 \times 10^9$.)
Electric potential and electric potential energy are scalar quantities, just like gravitational potential and gravitational potential energy. The work done is independent of the path followed.
The work done W in moving charge q from A to B is: $W = q\Delta V_e$, where ΔV_e is the potential difference between A and B.
Electric potential is a scalar quantity. Therefore, if we have i charges q_i with distance of r_i to the point P, then the electric potential at point P would be $V_e = k\sum \frac{q_i}{r_i}$.
For a point P outside the sphere and at a distance r from the centre of the sphere, the potential at P is $V_e = \frac{kQ}{r}$.
On the surface of the sphere the potential is $V_e = \frac{kQ}{R}$ where R is the radius of the sphere.
But at any point inside the sphere, the electric potential is constant and has the same value as the potential at the surface.

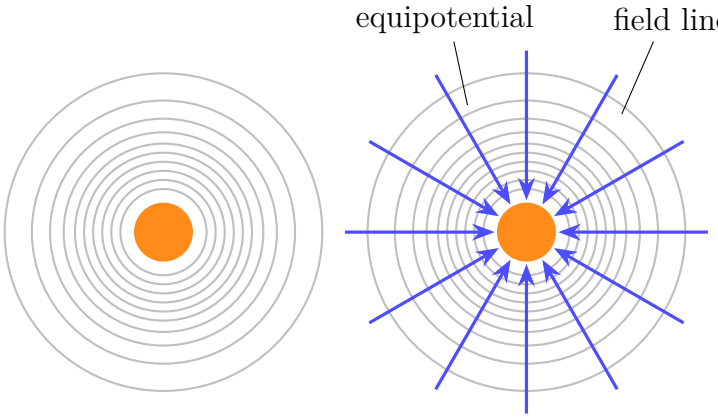
There is a deep connection between potential and field for both gravitation and electricity. To see this connection let us move a point charge q from a point P to a neighbouring point Q. There is a potential difference ΔV_e between the two points, and the points are a distance Δr apart.

$$W = qV_e = F \times d = qE\Delta r \Rightarrow E = \frac{\Delta V_e}{\Delta r}\left(-\frac{dV_e}{dr}\right)$$

In a graph showing the variation with distance of the potential, the gradient (slope) of the graph is the magnitude of the field strength. This applies to both gravitational and electric fields.

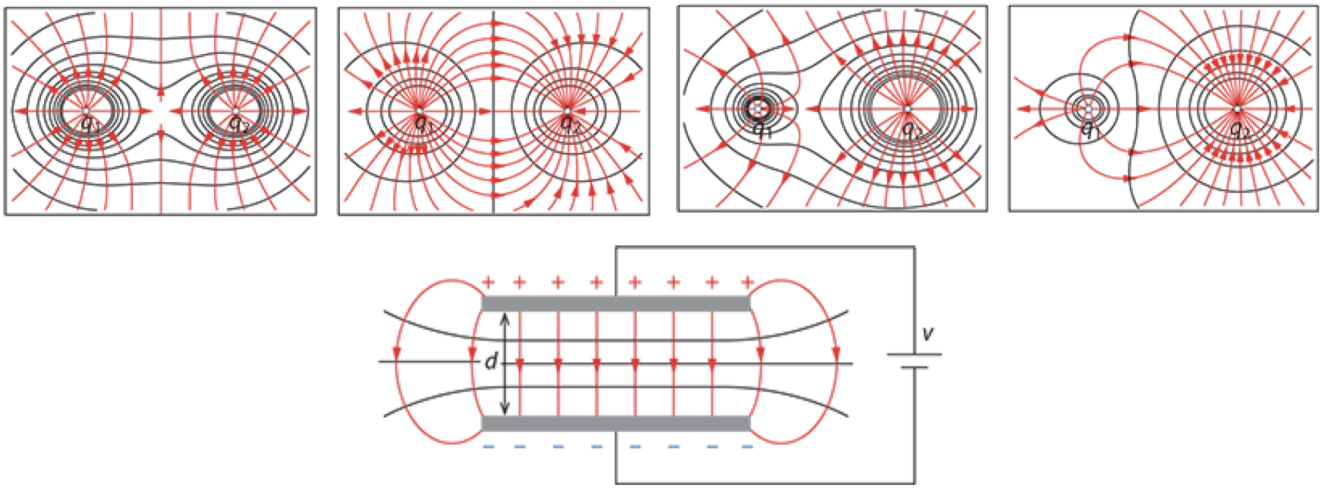
The connection between field lines and equipotential surfaces

Equipotential surfaces for a point charge.



The field lines are at right angles to the equipotential surfaces.

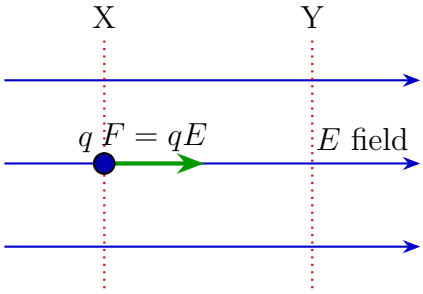
Explanation for why field lines are at right angles to the equipotential surfaces:
To move a charge from one point on an equipotential surface to another requires zero work. If the field lines were not normal to the equipotential surfaces, there would be a component of the field along the equipotential and so a force on the charge. As the charge moved this force would do work, which contradicts the fact that the work should be zero.

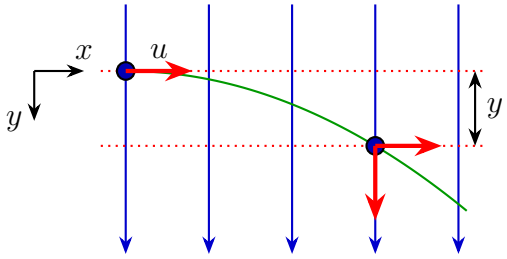


19 Motion in electric and magnetic fields

19.1 Motion in an electric field

The path of an electric charge in a uniform electric field is either a straight line or a parabolic path. We have a straight line if the charge is initially at rest or if its original velocity is parallel to the electric field. In all other cases we have a parabolic path.
A positive particle will experience a force qE in the direction of the field. It will also accelerate in the direction of the field, $a = \dfrac{qE}{m}$.
The positive charge increases its kinetic energy (= work done by eletric force, qV) as it moves to the right.





In the second figure:

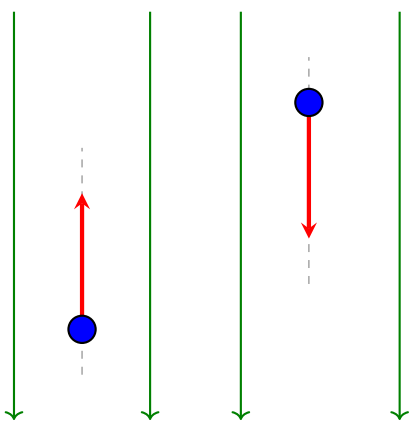
- the particle will follow a parabolic path
- the component of velocity normal to the field will remain constant: $v_x = u$
- the component of velocity parallel to the field will increase uniformly: $v_y = \dfrac{qE}{m}t$
- The kinetic energy of the particle increases as time goes on because the electric force is doing work on the particle.
- After time t, the particle will move a horizontal distance $x = ut$ and fall a vertical distance $y = \dfrac{1}{2}\dfrac{qE}{m}t^2$.

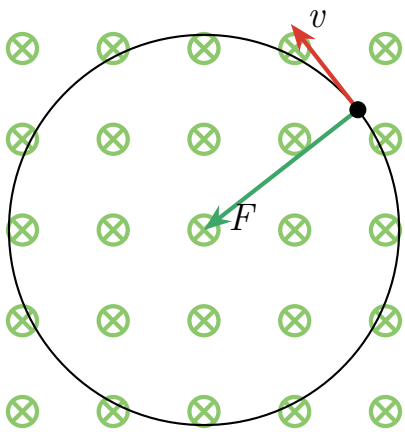
- $$\begin{aligned} E_K &= \frac{1}{2}mv^2 \\ &= \frac{1}{2}m\left(v_x^2 + v_y^2\right) \\ &= \frac{1}{2}m\left(u^2 + \left(\frac{qE}{m}t\right)^2\right) \end{aligned}$$
- $\therefore \Delta E_K = \frac{1}{2}m\left(\frac{qEt}{m}\right)^2 = Fy$, work done by the electric force.

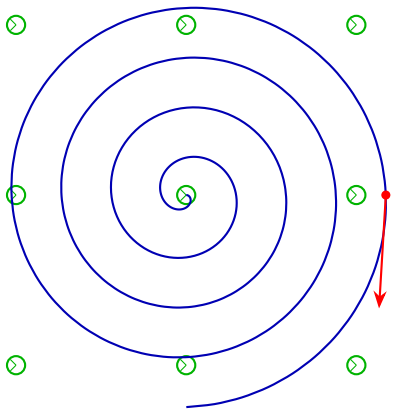
Equations involved in trajectory in Chapter 1 also applies here.

19.2 Motion in a magnetic field

- In the simple case where the velocity of the charged particle is parallel or anti-parallel to that of the uniform magnetic field, the magnetic force is zero and so the particle will move along a straight line.
- A positive charge in a uniform magnetic field that is directed into the plane of the paper. The velocity is at right angles to the magnetic field. The magnetic force is at right angles to both the velocity and the magnetic field so the particle will follow a circular path; the centripetal force is provided by the magnetic force.
  Radius $R = \frac{mv}{qB}$, time to make one complete revolution $T = \frac{2\pi m}{qB}$ (independent of the speed).
- If the positively charged particle loses energy as it moves along the circular path, the speed will decrease, and from the formula for the radius we see that the radius will also decrease. This means that the circular path will become a spiral.





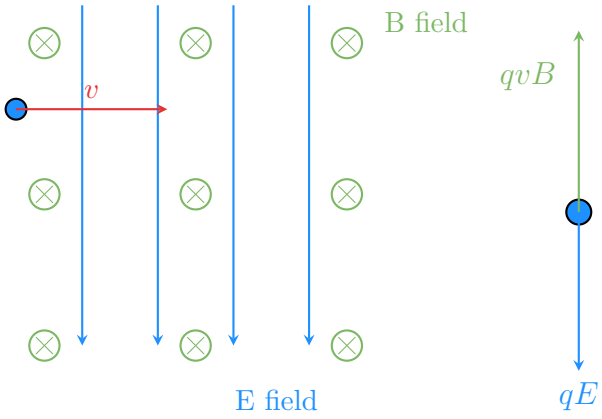


Work done and magnetic forces

Since the magnetic force is always normal to the velocity of the charge, it follows that it cannot do any work. This means the kinetic energy of the particle cannot change while in the region of magnetic field.

Motion in crossed electric and magnetic fields

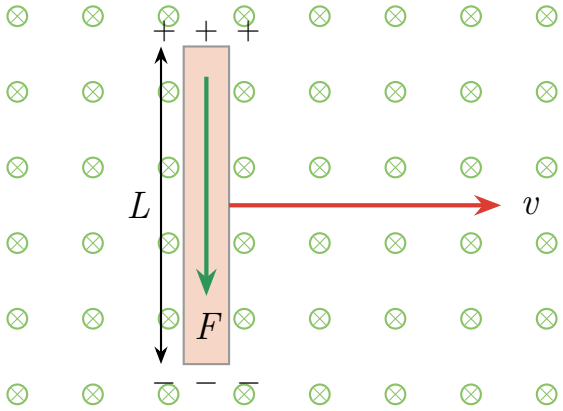
- If two forces are equal, the particle will continue moving on a straight line undeflected from its original path, $E = vB$.
- If the particle enters with speed greater than $v = \frac{E}{B}$, the particle will move upwards (larger F_m).
- If the speed is less than v, the particle will move downwards.



20 Electromagnetic induction

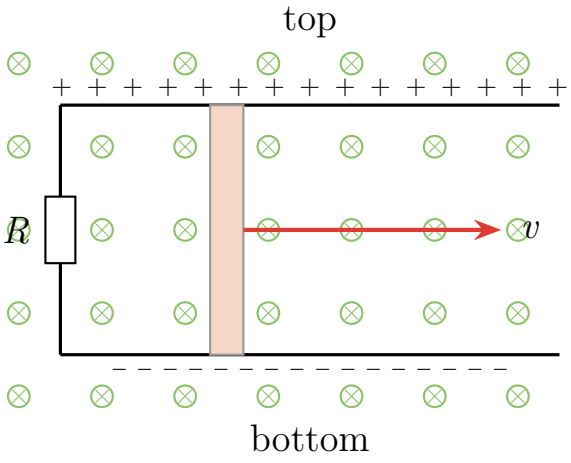
20.1 Electromagnetic induction

Motional *e.m.f*



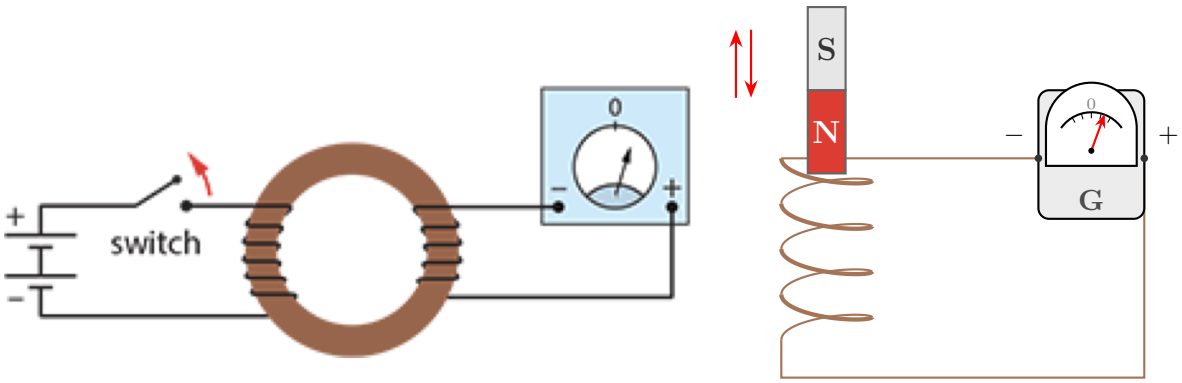
The rod is conducting. As it moves, the electrons within it also move from left to right. The magnetic field will exert a force on these moving electrons. The force on the electrons is directed downward. This means that the lower end of the rod has a net negative charge and the top end has an equal net positive charge. The flow of electrons towards the bottom end of the rod will stop when the electrons already there are numerous enough to push any new electrons back by electrostatic repulsion. There is, in other words, an electric field established in the rod whose direction is from top to bottom.

$E = \frac{\varepsilon}{L}$, where ε is the induced *e.m.f* and L is the length of the rod. The induced *e.m.f* is given by $\varepsilon = BLv$. It is important to note that except for a very short interval of time initially, no current exists in the rod.



Now there exists a current of $I = \dfrac{BvL}{R}$ in the circuit, and the moving rod behaves as a battery. In this diagram the current is going counterclockwise.
Notice also that now that we have a current, the rod needs to be pushed if it is to continue to move at constant speed. Now it's experiencing a magnetic force of $F = \dfrac{vB^2L^2}{R}$ directed to the left.
For speed to remain constant a force of equal magnitude needs to act on the rod, directed to the right. The power P generated by this force $= \dfrac{(BvL)^2}{R}$, consistent with the power dissipated in the circuit as thermal energy.

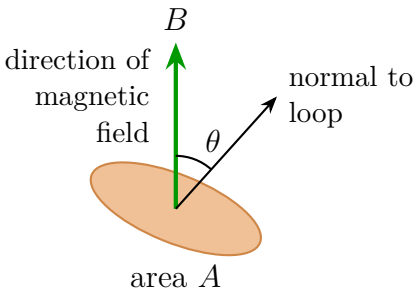
Magnetic flux and magnetic linkage



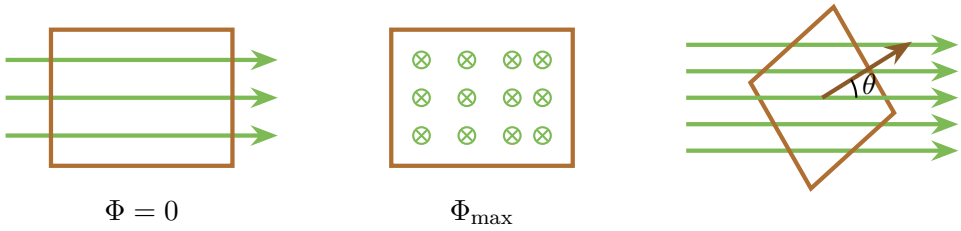
The current is created as a result of the motion of the magnet relative to the coil. The current registered by the galvanometer increases when:

- the relative speed of the magnet and the coil increases
- the strength of the magnet increases
- the number of turns increases
- the area of the loop increases
- the magnet moves at right angles to the plane of the loop.

The magnetic flux through the loop of wire is given as $\Phi = BA\cos\theta$, where B is the magnetic flux density, A is the area of the loop and θ is the angle between the normal to the plane of the loop and the direction of the magnetic field.



The magnetic flux multiplied by the number of turns in a loop is called the flux linkage, $\Phi = NBA\cos\theta$.
The intuitive picture of magnetic flux is the number of magnetic field lines that pierce the loop area.



Faraday's Law

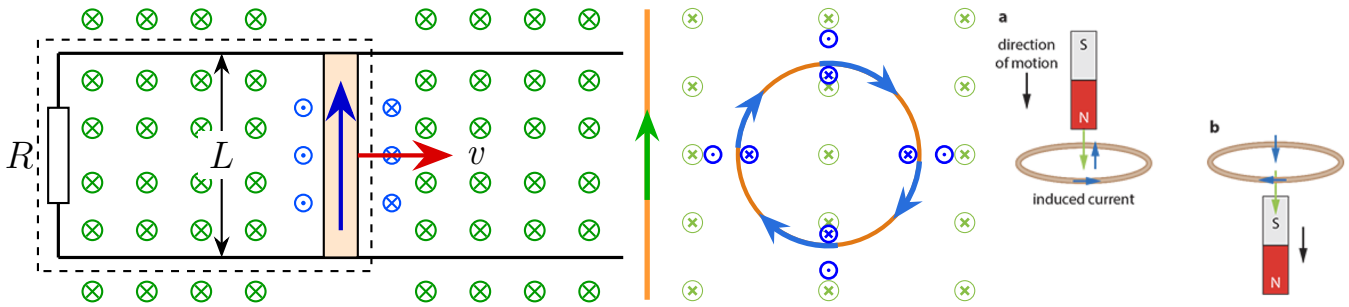
The current generated by a magnetic field is correlated with the rate of change of magnetic flux linkage through the loop. Changing flux creates an induced $e.m.f$, not necessarily a current. There will only be a current if the loop is conducting; that is, if the resistance of the circuit is not infinite. The induced $e.m.f$ is equal to the (negative) rate of change of magnetic flux linkage:

$$\varepsilon = -N\frac{\Delta\Phi}{\Delta t}\left(= -N\frac{\mathrm{d}\Phi}{\mathrm{d}t}\right)$$

Lenz's Law

It states that the direction of the induced current is such that it opposes the change in magnetic flux that produced it. This is a consequence of the conservation of energy and is reflected in the negative sign in Faraday's Law.

- If the flux increases, the induced $e.m.f$ will create a current whose magnetic field is opposite to the external field.
- If the flux decreases, the induced $e.m.f$ will create a current whose magnetic field is parallel to the external field.



- The rod is made to move to the right. The magnetic flux through the loop is increasing and a current will be established in the rod. By Lenz's law, the current must produce a magnetic field opposite to the existing magnetic field, i.e. out of the page. (Opposing the increase of the magnetic flux ⇒ decrease the magnetic flux, and one way is to set up a magnetic field inside the loop of the opposite direction) So the current must be counterclockwise.
- A loop of wire near a straight wire in which the current is decreasing. The current in the straight wire creates a magnetic field into the page at the position of the loop. The induced current in the loop produces a magnetic field in the same direction as to oppose the change in flux. From the right-hand rule, the current must be clockwise.
- The flux in the loop is increasing because the magnetic field at the loop is getting larger as the magnet approaches. The induced current must then oppose the increase in the flux. This can be done if the induced current produces a magnetic field in the opposite direction to that of the bar magnet. Thus, the current will flow in a counterclockwise direction when looked at from above.

Self induction

By changing the resistance of R the current in the circuit will be changing; the magnetic flux linkage in the coil will therefore also be changing and an $e.m.f$ ε_L will be induced in the coil. By Lenz's law, the induced $e.m.f$ will oppose the change in the current. The polarity of the induced $e.m.f$ is such as to drive current in the opposite direction to that of the current from the cell. In this way the induced $e.m.f$ opposes the increase in the current by decreasing the current in the circuit.

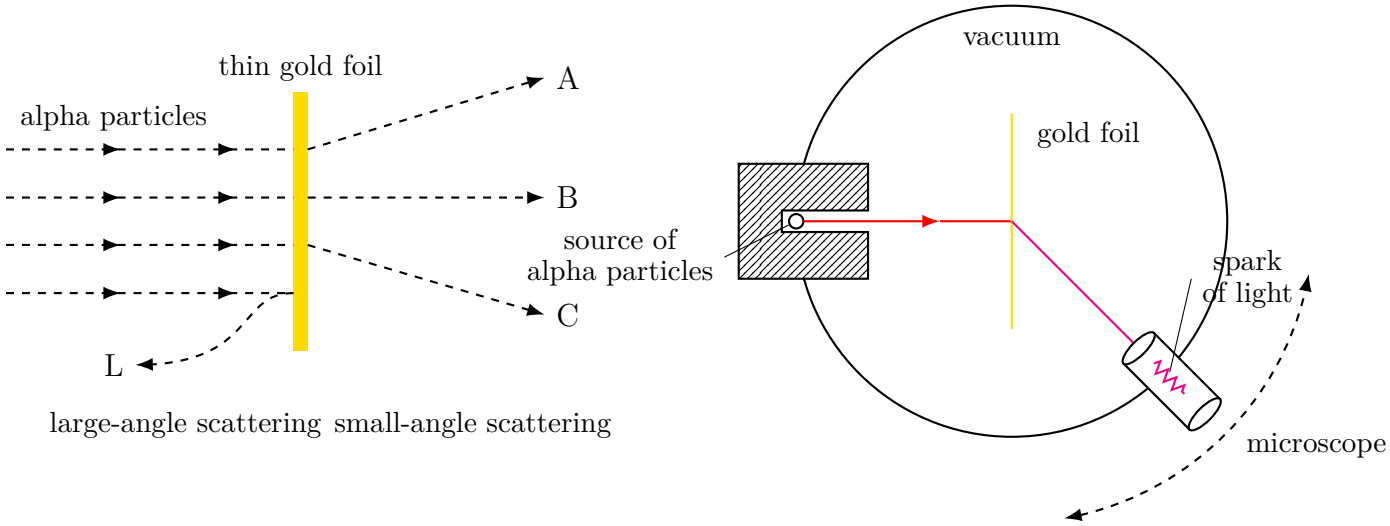
Unit E Nuclear and quantum physics

21 Atomic physics

21.1 The structure of the atom

α particles were directed at a thin gold foil in an evacuated chamber. 你Rutherford哥哥又限时返场了。They found that:

- the great majority of the alpha particles went straight through the foil with no or little deflection; they were detected at very small scattering angles: they were due to the electric force of repulsion between the positive charge of the gold atoms and the positive charge of the alpha particles. The effect of the electrons of the gold atoms on the path of the alpha particles is negligible.
- very occasionally, alpha particles were detected at very large scattering angles. **The repulsion force must be very large and thus the separation must be very small. This could only be explained if the positive charge on the atom is within a sphere that is much smaller than the sphere that Thomson had imagined.** This and subsequent experiments confirmed the existence of the nucleus inside the atom: a small, massive object carrying the positive charge of the atom and most of its mass.



结论: The Rutherford experiment established the Rutherford model of the atom: the atom consists of a very small, massive, positively charged nucleus, and the electrons orbit the nucleus like planets around the sun.

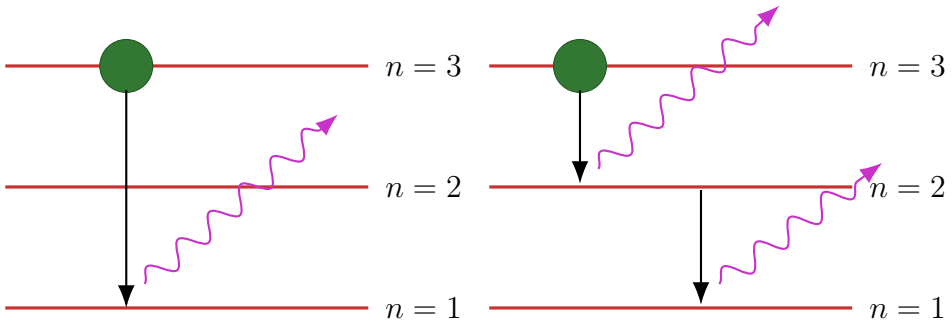
Discrete energy

If you expose a container of gas at low pressure to a strong electric field, light will be emitted from the gas. You can show that the emitted light contains many different wavelengths by passing it through a prism or diffraction grating. The result is a series of bands of light at different wavelengths. The set of possible wavelengths that can be emitted is called the emission spectrum.

Bohr argued that an atom has discrete energy, i.e. it can have one out of a specific set of values. He represented the possible energies (refers to $E_T = E_K + E_P$) with an energy level diagram. For example, the hydrogen atom in the nth state has an energy of $E = -\dfrac{13.6}{n^2}$ eV, $n \in \mathbb{Z}^+$. The negative sign in the energy tells us that the electron is trapped within the atom. To have the electron leave the atom, energy must be provided.
Bohr suggested that an atom can make a transition from a state of higher energy to a state of lower energy by emitting a photon. The energy of the emitted photon is the difference in energy of the two levels.

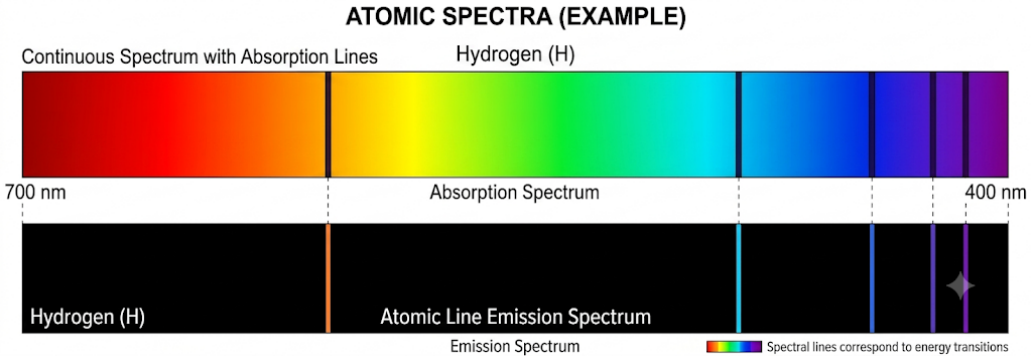
The energy of a photon is given by $E = hf = h\frac{c}{\lambda}$, where $h = 6.63 \times 10^{-34}$ J s is Planck's constant.

In a transition from a high to lower energy state, the wavelength of the emitted photon is given by $\lambda = \frac{hc}{\Delta E}$.
When undisturbed, the electron in the atom will occupy the lowest energy state, called the ground state. When energy is supplied to the atom, the electron can be excited to a higher energy state, called an excited state. From the excited state, the electron can return to the ground state by emitting photons, either going to ground state straightaway or passing through multiple transition energy levels, and the decision between the two chances is an issue of chance.



Absorption Spectra

Now imagine sending a beam of white light through a gas. The majority of the atoms in the gas are in their ground state. Electrons in the atoms may absorb photons in the beam and move to an excited state. This will happen only if the photon that is to be absorbed has exactly the right energy. This means that the light that is transmitted through the gas will be missing the photons that have been absorbed. This gives rise to absorption spectra.
Note: Why are the photons missing? This is because the photons are emitted in all directions and not necessarily along the direction the observer is looking.



Experiments show that no two elements have the same wavelengths in their emission and absorption spectra.

21.2 Quantisation of angular momentum

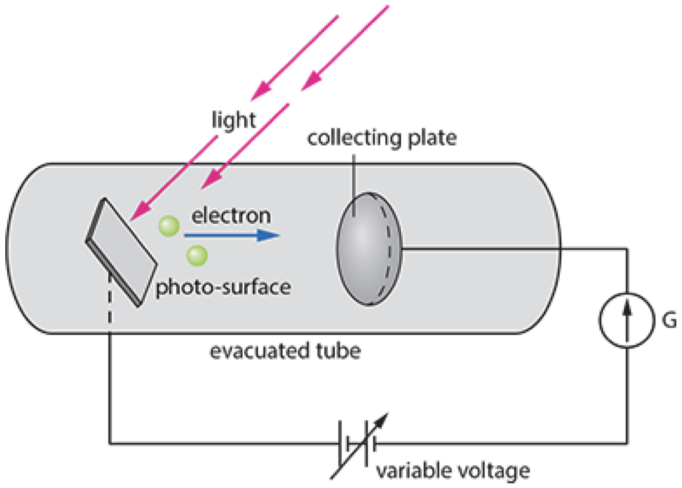
Bohr assumed that the electron could only exist in certain special orbits satisfying the condition

$$mvr = n\frac{h}{2\pi}, \ n \in \mathbb{Z}^+$$

where $L = mvr$, angular momentum of the electron, is quantised. The electron can only occupy orbits for which the angular momentum is an integer multiple of $\frac{h}{2\pi}$.

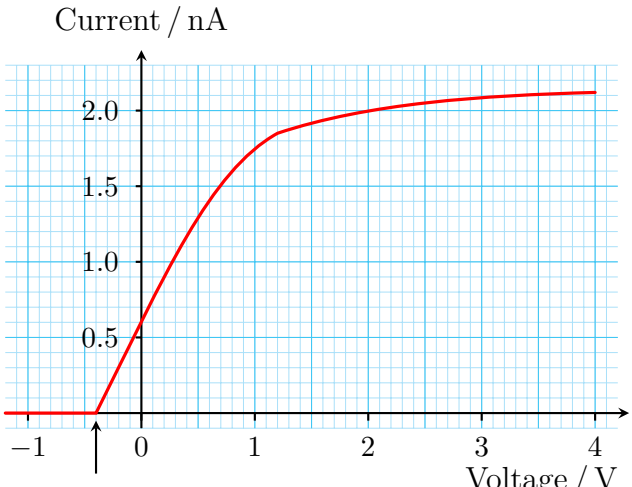
Photoelectric effect

The photoelectric effect is the phenomenon in which light (or other forms of electromagnetic radiation) incident on a metallic surface causes electrons to be emitted from the surface.

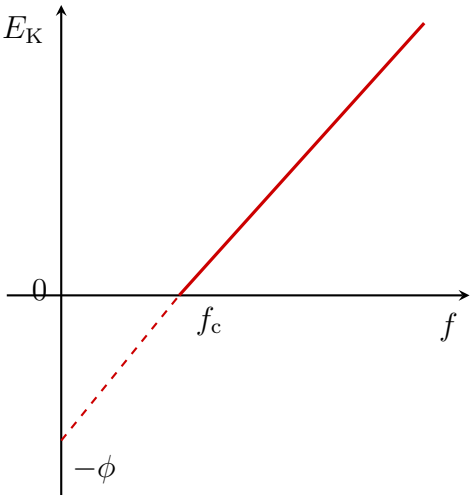


The emitted electrons are called photoelectrons. Those electrons that make it to the collecting plate complete the circuit and so we have an electric current that is recorded by the sensitive galvanometer.

- Note that the terminal of the variable power supply connected to the collecting plate is the negative terminal. This means that the collecting plate actually repels the emitted electrons. Only the very energetic electrons will make it to the plate. As the magnitude of the voltage is increased (i.e. made more negative) fewer and fewer electrons make it to the plate; eventually no electron will arrive there and at that point the current becomes zero. The voltage at which the current becomes zero is called the stopping voltage V_s, which constitutes the formula for maximum kinetic energy for emitted photons, $E_{\max} = eV_s$.
- Now the positive terminal is connected to the collecting plate. The electrons are now attracted to the collecting plate and the current increases. As the voltage is increased even more the current saturates, i.e. it approaches a constant value. This is because the collecting plate is so positive that it attracts every single emitted electron.



the stopping voltage is this number taken as positive



Several conclusions were made:

1. Changing the intensity of light does not affect the stopping voltage, they release electrons of the same kinetic energy.

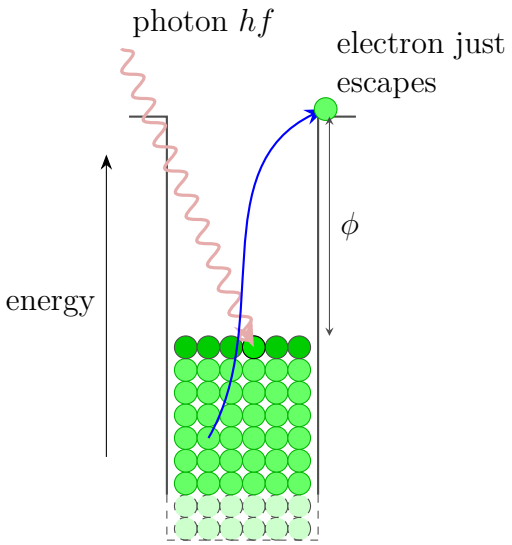
2. As f of light increases, $E_{\max}$ and V_s increases as well.

3. If we plot the kinetic energy of the electrons (which equals eV_s) versus frequency, we find a straight line. There exists a critical / threshold frequency f_c such that no electrons at all are emitted if the frequency of the light source is less than f_c. The frequency varies with different photo-surfaces.

4. Electrons are emitted with no time delay.

Einstein's explanation

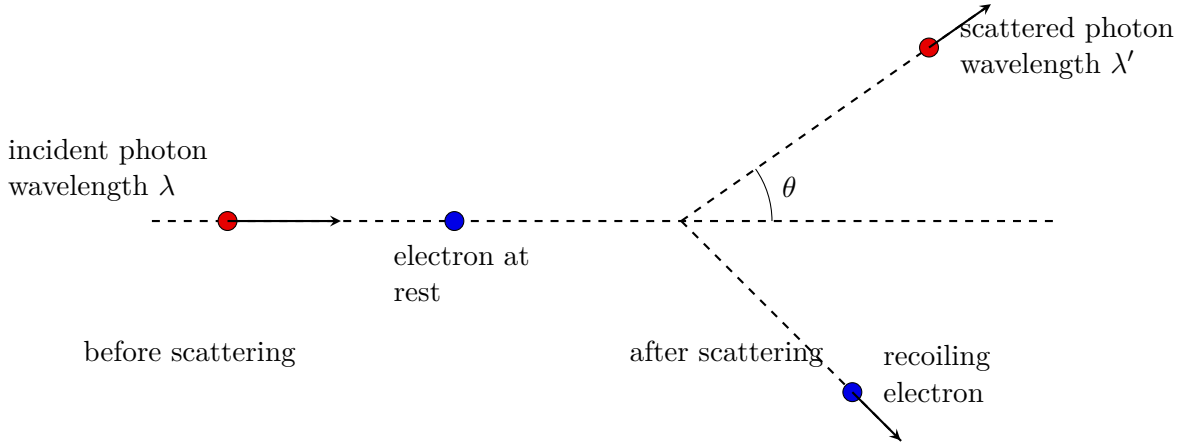
A single photon of frequency f is absorbed by a single electron in the photo-surface, and so the electron's energy increases by hf. The electron will have to spend a certain amount of energy, ϕ (the work function, given by $\phi = hf_c = hf - eV_s$), to free itself from the pull of the nuclei of the atoms of the photo-surface. The electron will be emitted (become free) if hf is bigger than ϕ. The difference $E_K = hf - \phi$ will be the kinetic energy of the (now) free electron.



We are thus able to derive this formula for stopping voltage: $V_s = \frac{h}{e}f - \frac{\phi}{e}$.

Compton scattering

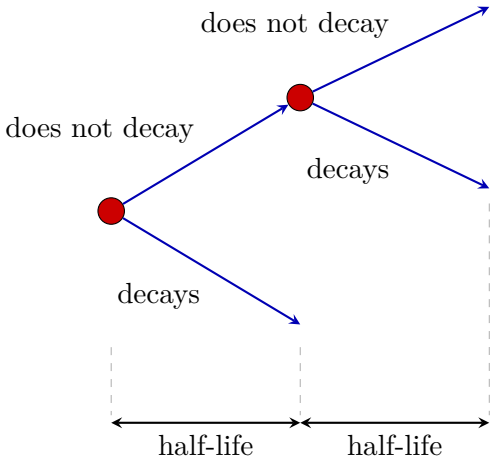
This is the scattering of light off an electron. The scattered light bounces off the electron with a longer wavelength. We can understand this in terms of photons easily: the light gives some energy to the electron which recoils with kinetic energy. The energy of the light has therefore been reduced.

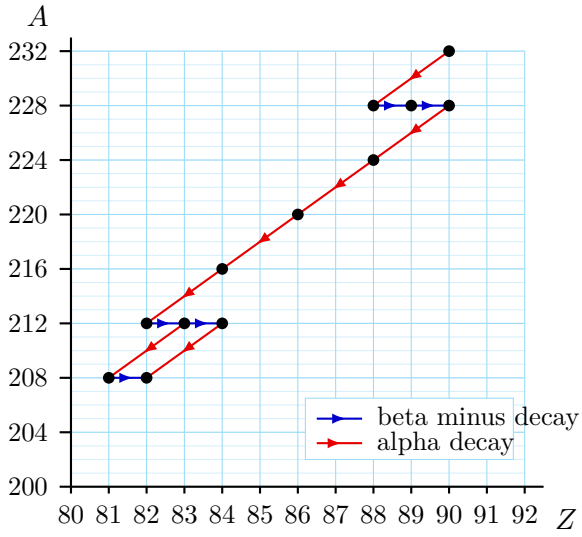


有人注意到这个图其实之前出现过么?

Half-life and probability

The meaning of a half-life can also be understood in terms of probability. Any given nucleus has a 50% chance of decaying within a time interval equal to the half-life. If a half-life goes by and the nucleus has not decayed, the chance that it will decay in the next half-life is still 50%.



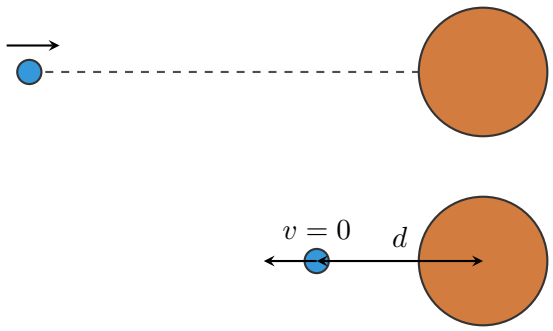


23.3 Nuclear properties and the radioactive decay law

Nuclear radii

When an alpha particle of kinetic energy E_K is directed at a nucleus of proton number Z, the distance of closest approach, d, is found by equating the kinetic energy far from the nucleus to the electric potential energy at the point of closest approach.

$$E_K = k\frac{(2e)(Ze)}{d} \Rightarrow d = \frac{2kZe^2}{E_K}$$



As the energy of the incoming alpha particles increases, the distance of closest approach decreases and will eventually approach the nuclear surface.

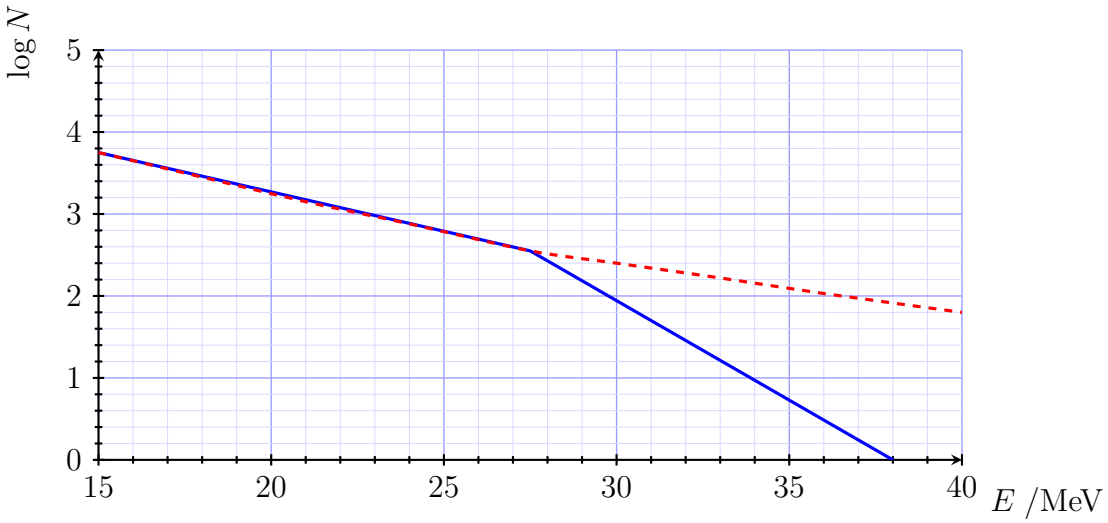
When the alpha particle is near the nuclear surface deviations from Rutherford scattering will be observed because the nuclear force will start acting on the alpha particle. The number of scattered alpha particles will not be what is expected based on Rutherford scattering. Thus, the distance of closest approach measured in this case will essentially be the nuclear radius.

The radius of a nucleus is given by $R = R_0 A^{1/3}$, where $R_0 \approx 1.2 \times 10^{-15}$ m and A is the nucleon number. (This formula also implies that the volume of a nucleus is proportional to the number of nucleons in it, and that the density of nuclei is constant, $\rho = 2 \times 10^{17}$ kg m^{-3}.)

Deviations from Rutherford scattering — evidence for the strong nuclear force

The Rutherford formula states that as the scattering angle increases, the number of alpha particles scattered at that angle decreases very sharply. It states that the number N of alpha particles scattering at an angle θ is proportional to $\frac{1}{\sin^4(\theta/2)}$. (If this was the case, $N \sin^4\left(\frac{\theta}{2}\right)$ would be a constant for all angles of θ.)

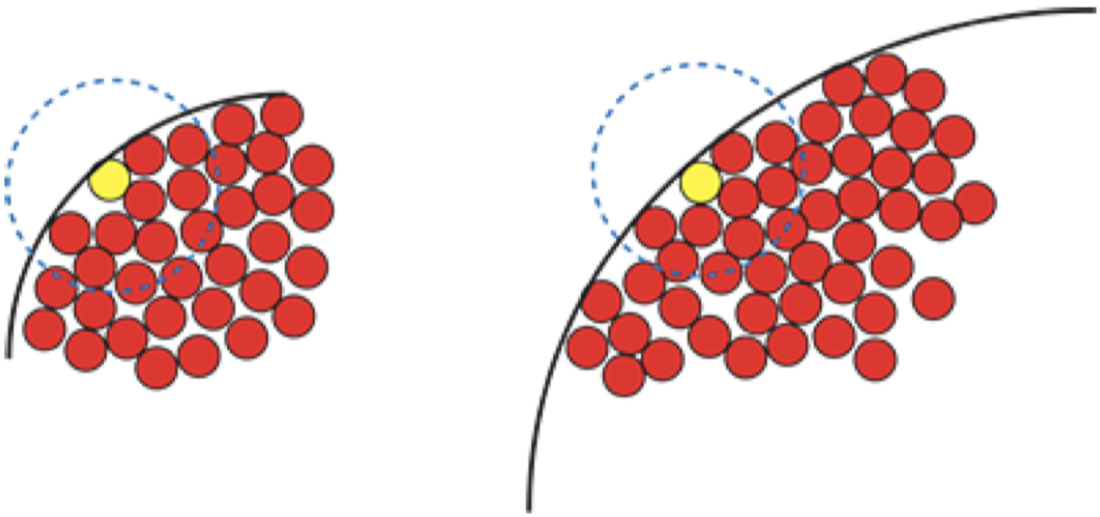
The derivation of the Rutherford formula is based on a number of assumptions. The most important is that the only force in play during the scattering process is the electric force. As the energy of the alpha particles increases, the alpha particles can get closer to the nucleus. When the distance of closest approach gets to be about 10^{-15} m or less, deviations from the Rutherford formula are observed. This is due to the fact that the alpha particles are so close to the nucleus that the strong nuclear force begins to act on the alpha particles.



The dotted curve is based on Rutherford scattering. The blue curve is the observed curve. We see deviations when the energy exceeds about 28 MeV. The energy at which deviations start may be used to estimate the nuclear radius.

Approximate constancy of the binding energy per nucleon

The short range of the force implies that a given nucleon can interact with its immediate neighbours only and not with all of the nucleons in the nucleus. So, for large nuclei (roughly A > 20) any one nucleon is surrounded by the same number of immediate neighbours, and so the energy needed to remove that nucleon from the nucleus is the same.



- **White dwarfs**
  - The bottom left is a region of small stars (hot).
  - These are common stars, but their faintness makes them hard to detect.

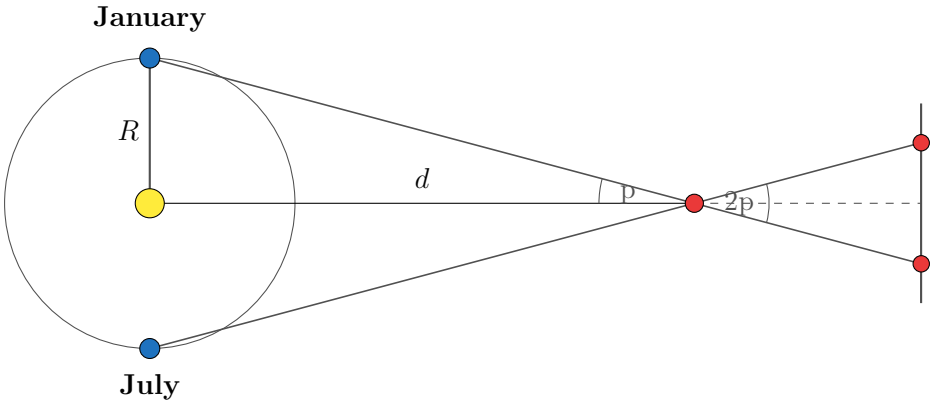
Type of star	**Main characteristic**
Main sequence star	Fuses Hydrogen into Helium
Red giant	Bright, large, cool, reddish, tenuous
Super red giant	Even larger and brighter than red giants
White dwarf	Dim, small, hot, whitish, dense
Neutron star	Dimmer and smaller than a white dwarf with a density similar to that of nuclei. Too dim to show on an HR diagram
Black hole	The end point of the very massive stars

The stellar radii can be given by $R = \sqrt{\dfrac{L}{4\pi\sigma T^4}}$.

To measure the radius of a star through its temperature and luminosity assumes we know the temperature and luminosity! Temperature can be measured using Wien's law, while luminosity can be found by apparent brightness and distance.

Parallax methods

It can be used to measure the distance. It is based on the fact that when you look at an object from two different positions, the object appears to shift relative to the very distant background. The method is restricted to nearby stars. Stars far away have parallax angles that are too small to measure accurately. The parallax angle is the angle at which the star subtends a distance equal to the earth orbit radius R.



Parallax angles are very small. It is convenient to express them in arc seconds: $1^\circ = 60$ arcmin $= 3600$ arcsec. One parsec is the distance at which the parallax is 1 arcsec, $d(/\text{pc}) = \dfrac{1}{p(/\text{arcsec})}$, 1 pc $= \dfrac{R}{\tan p} = \dfrac{1.5 \times 10^{11}}{\tan(1/3600)^\circ} \approx 3.09 \times 10^{16}$ m $= 3.27$ ly.

The proton-proton chain

The main series of nuclear fusion reactions taking place in the cores of low mass main sequence stars is the proton-proton chain:

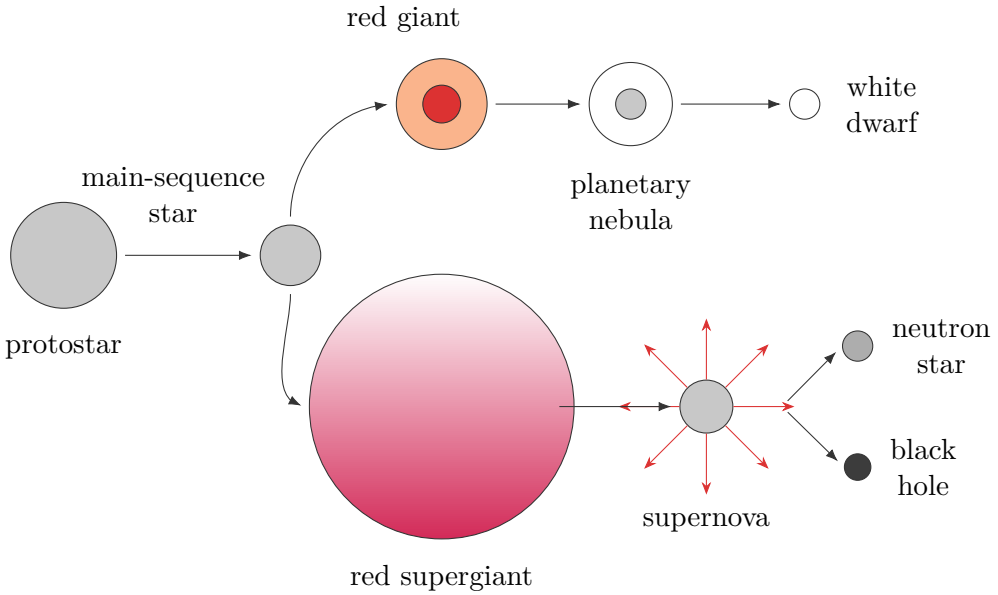
$$
\begin{aligned}
{}^1_1\mathrm{H} + {}^1_1\mathrm{H} &\rightarrow {}^2_1\mathrm{H} + {}^0_1 e^+ + \nu \\
{}^1_1\mathrm{H} + {}^2_1\mathrm{H} &\rightarrow {}^3_2\mathrm{He} + \gamma \\
{}^3_2\mathrm{He} + {}^3_2\mathrm{He} &\rightarrow {}^4_2\mathrm{He} + 2{}^1_1\mathrm{H}
\end{aligned}
$$

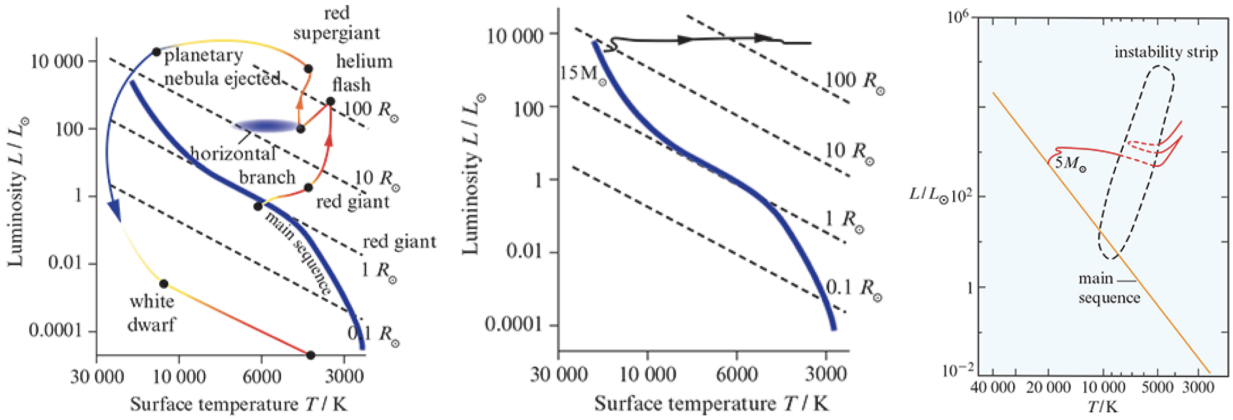
In this cycle, the net effect is to turn four hydrogen nuclei into one nucleus of helium.

$$4{}^1_1\mathrm{H} \rightarrow {}^4_2\mathrm{He} + 2{}^0_1 e^+ + 2\nu + 2\gamma$$

Stellar evolution

When about 12% of the hydrogen in the star is used up in nuclear fusion, a series of instabilities develop in the star. These have the result that the delicate balance between the radiation and gravitational pressures is upset. The star will then begin to move away from the main sequence. Other types of nuclear fusion reactions will take place and the star changes in size, surface temperature, and color.



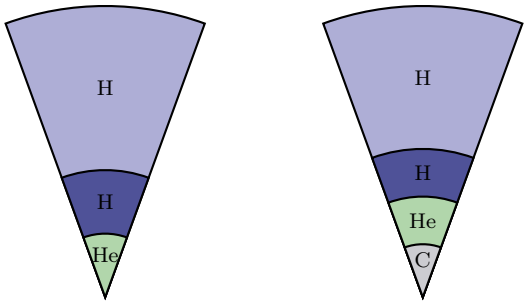


A star more massive than the sun may go back and forth on the HR diagram several times. The delicate balance between radiation and gravitational pressures has been disturbed; the star tries to find a new equilibrium state but instead the star pulsates, i.e. it grows and then shrinks in size. So the star cools and heats up during these pulsations and that affects the rate and type of nuclear fusion reactions taking place. This makes its luminosity variable, and this is a general characteristic of all stars in the instability region.

- In more massive stars, carbon will replace helium in the core once carbon starts getting produced in the triple alpha process:

$$^4_2\text{He} +^4_2 \text{He} \rightarrow^8_4 \text{Be} + \gamma$$
$$^4_2\text{He} +^8_4 \text{Be} \rightarrow^{12}_6 \text{C} + \gamma$$

  - Only hydrogen in the thin shell is undergoing nuclear fusion.
  - The temperature and pressure of helium build up and eventually helium begins to fuse through the triple alpha process, called the "Helium flash".
  - Carbon collects in the core, with helium now fusing in a thin shell around carbon.
  - This release of energy blows away the outer layers of the star in an explosion called a planetary nebula; mass is thrown into space leaving behind a helium or carbon core.



For stars with different sizes of mass:

- $M < 0.25M_\odot$: white dwarf with Helium core:

$$4^1_1\text{H} \rightarrow^4_2 \text{He} + 2^0_1 e^+ + 2\nu + 2\gamma$$

- $0.25M_\odot < M < 8M_\odot$: white dwarf with Carbon core (triple alpha process just mentioned)
- $8M_\odot < M < 12M_\odot$: white dwarf with Oxygen / Neon / Magnesium core:

$$^4_2\text{He} +^{12}_6 \text{C} \rightarrow^{16}_8 \text{O} + \gamma$$
$$^{12}_6\text{C} +^{12}_6 \text{C} \rightarrow^{20}_{10} \text{Ne} +^4_2 \text{He}$$
$$^{12}_6\text{C} +^{12}_6 \text{C} \rightarrow^{23}_{11} \text{Na} +^2_1 \text{H}$$
$$^{12}_6\text{C} +^{12}_6 \text{C} \rightarrow^{23}_{12} \text{Mg} +^1_0 n$$
$$^{16}_8\text{O} +^{16}_8 \text{O} \rightarrow^{28}_{14} \text{Si} +^4_2 \text{He}$$

- $12M_\odot < M < 40M_\odot$: Neutron star
- $M > 40M_\odot$: Black hole

Fusion cannot produce elements heavier than iron, since the binding energy per nucleon peaks near iron and further fusion is not energetically possible. Elements heavier than iron are produced during supernova explosions through neutron absorption.

